

# Physics-Informed Federated Learning Framework for Electric Vehicle Energy Consumption Prediction with Real-Time Traffic Simulation Using SUMO

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## Abstract

The rapid adoption of electric vehicles (EVs) has intensified the need for accurate energy consumption prediction models to support intelligent transportation systems, battery management, route planning, and charging infrastructure optimization. Traditional data-driven approaches often rely solely on historical vehicle telemetry and overlook the underlying physical principles governing vehicle dynamics. Furthermore, centralized machine learning frameworks require the collection of raw data from multiple sources, raising concerns regarding privacy, scalability, and data ownership. To address these limitations, this study proposes a Physics-Informed Federated Learning Framework for Electric Vehicle Energy Consumption Prediction with Real-Time Traffic Simulation.

The proposed framework integrates vehicle telemetry, battery characteristics, environmental conditions, and road attributes with physics-based power estimation features derived from aerodynamic drag, rolling resistance, grade resistance, and inertial forces. These engineered features are combined with a Light Gradient Boosting Machine (LightGBM) regression model to improve predictive accuracy. A federated learning architecture is further incorporated to enable privacy-preserving distributed model training across multiple vehicle groups without requiring the exchange of raw data. The experimental dataset consists of 5,000 electric vehicle operational records containing driving behavior, battery parameters, weather conditions, and trip-level information. To evaluate the contribution of different feature sets, four predictive configurations are investigated, including raw-feature learning, reduced-feature learning, physics-only learning, and hybrid physics-informed learning.

Experimental results demonstrate that the proposed hybrid physics-informed model achieves the best performance with a Mean Absolute Error (MAE) of 0.4629 kWh, a Root Mean Square Error (RMSE) of 0.5744 kWh, and an  $R^2$  score of 0.9320, significantly outperforming physics-only and reduced-feature baselines. The federated implementation achieves a comparable MAE of 0.4844 kWh, indicating that privacy-preserving distributed learning can be achieved with minimal loss in predictive performance. To demonstrate real-world applicability, the trained model is integrated with the Simulation of Urban Mobility (SUMO) platform for real-time vehicle energy prediction and visualization. An interactive analytics dashboard is also developed to provide model evaluation, performance monitoring, and vehicle-level insights.

The findings indicate that combining physics-informed feature engineering with federated machine learning offers an effective and scalable solution for accurate EV energy consumption prediction while maintaining data privacy. The proposed framework provides a practical foundation for next-generation intelligent transportation systems, smart charging infrastructure, and large-scale EV fleet management applications.

**Keywords**— Electric Vehicle Energy Consumption Prediction, Physics-Informed Machine Learning, Federated Learning, LightGBM, Intelligent Transportation Systems, SUMO Traffic Simulation, Privacy-Preserving Learning, Energy Analytics.

## I. INTRODUCTION

### A. Background

The global transportation sector is undergoing a significant transformation driven by the rapid adoption of electric vehicles (EVs). Growing concerns regarding greenhouse gas emissions, climate change, fossil fuel dependency, and urban air pollution have accelerated the transition from conventional internal combustion engine vehicles to electrified transportation systems. Governments, industries, and research organizations worldwide are actively promoting EV deployment through regulatory policies, financial incentives, and investments in charging infrastructure. Consequently, electric vehicles have emerged as a key technological solution for achieving sustainable mobility and reducing the environmental impact of transportation systems.

Despite these advantages, several operational challenges continue to hinder large-scale EV adoption. One of the most critical challenges is the accurate estimation and prediction of vehicle energy consumption. Unlike conventional vehicles, whose refueling process is relatively fast and predictable, EV performance is strongly influenced by battery capacity, driving behavior, traffic conditions, environmental factors, road characteristics, and vehicle dynamics. Inaccurate estimation of energy consumption may lead to range anxiety, inefficient charging schedules, suboptimal route planning, and increased operational costs for fleet operators.

Accurate energy consumption prediction plays a vital role in numerous EV-related applications, including battery management systems, intelligent navigation, charging station recommendation systems, fleet energy optimization, and smart grid integration. Reliable prediction models enable vehicle operators and transportation authorities to estimate future energy requirements, optimize charging infrastructure utilization, and improve overall transportation efficiency. Consequently, the development of

robust and accurate EV energy prediction frameworks has become an active research area within intelligent transportation systems and sustainable energy management.

### *B. Challenges in EV Energy Consumption Prediction*

Predicting EV energy consumption is a complex problem due to the multidimensional interactions among vehicle, driver, environmental, and infrastructure-related factors. Energy consumption is influenced by vehicle speed, acceleration patterns, road gradients, battery characteristics, ambient temperature, wind conditions, traffic congestion, tire pressure, and vehicle mass. These variables often exhibit nonlinear relationships that are difficult to capture using traditional analytical approaches.

Conventional physics-based models estimate energy consumption using vehicle dynamics equations and power balance formulations. Although these models provide interpretability and physical consistency, they often require accurate parameter calibration and may struggle to represent complex real-world driving conditions. In contrast, data-driven machine learning approaches can capture nonlinear relationships directly from historical data but frequently lack physical interpretability and may exhibit poor generalization outside the observed data distribution.

Another major challenge arises from data privacy and ownership concerns. Modern EVs continuously generate large volumes of telemetry data containing detailed information about driving behavior, battery usage, and travel patterns. Traditional centralized machine learning approaches require collecting and aggregating this data in a central repository, which introduces concerns related to privacy, cybersecurity, data governance, and regulatory compliance. These challenges become particularly significant in large-scale fleet management systems where data originates from multiple geographically distributed sources.

Therefore, there is a growing need for predictive frameworks that simultaneously provide high prediction accuracy, physical interpretability, scalability, and privacy preservation.

### *C. Evolution of Data-Driven EV Energy Prediction*

Recent advancements in machine learning have significantly improved the accuracy of EV energy consumption prediction models. Various regression algorithms, including Random Forest, Support Vector Regression, Gradient Boosting Machines, XGBoost, and Light Gradient Boosting Machine (LightGBM), have demonstrated strong predictive capabilities across diverse transportation datasets. Deep learning techniques such as Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), Convolutional Neural Networks (CNN), and Transformer-based architectures have further expanded the ability to model complex temporal dependencies in vehicle operational data.

While these approaches have achieved promising results, many existing studies rely exclusively on raw sensor measurements and historical observations. Such models often treat the prediction process as a purely statistical problem, ignoring the fundamental physical mechanisms that govern vehicle energy consumption. As a result, prediction performance may degrade when models encounter driving scenarios that differ from those represented in the training dataset.

To address these limitations, researchers have increasingly explored Physics-Informed Machine Learning (PIML), which integrates domain knowledge and physical equations into data-driven learning frameworks. By incorporating physically meaningful variables such as aerodynamic drag, rolling resistance, grade resistance, and inertial power, PIML models can improve prediction accuracy while maintaining consistency with established vehicle dynamics principles.

### *D. Federated Learning for Privacy-Preserving Transportation Analytics*

Federated Learning (FL) has emerged as a promising paradigm for distributed machine learning in privacy-sensitive environments. Unlike centralized learning approaches, federated learning enables model training across multiple clients without requiring the exchange of raw data. Instead, each client performs local training and shares model updates or predictions, thereby preserving data privacy while facilitating collaborative learning.

The application of federated learning within intelligent transportation systems has gained considerable attention in recent years. Vehicle manufacturers, fleet operators, and transportation agencies increasingly seek methods for leveraging distributed data sources while complying with privacy regulations and organizational data-sharing constraints. Federated learning offers a practical solution by enabling decentralized model development across geographically distributed vehicles and transportation networks.

Despite the growing interest in federated learning, relatively few studies have investigated its integration with physics-informed EV energy prediction frameworks. Existing research often focuses on either privacy-preserving learning or energy prediction independently, leaving a significant opportunity to combine these complementary approaches into a unified framework.

### *E. Research Gap*

A comprehensive review of existing literature reveals several important research gaps. First, many EV energy prediction studies rely solely on raw telemetry data and do not incorporate explicit physical knowledge regarding vehicle energy dynamics.

Second, physics-based approaches often lack the flexibility required to model complex nonlinear interactions present in real-world transportation systems. Third, most existing machine learning frameworks adopt centralized training architectures that require the collection of sensitive vehicle data, raising privacy and scalability concerns.

Furthermore, only a limited number of studies have explored the combined use of physics-informed feature engineering and federated learning for EV energy prediction. The integration of these techniques with real-time traffic simulation environments remains largely underexplored. As a result, there is a need for a unified framework capable of delivering accurate predictions, preserving privacy, incorporating physical interpretability, and supporting real-time deployment within transportation simulation platforms.

#### F. Research Objectives

The primary objective of this research is to develop a physics-informed federated learning framework for accurate and privacy-preserving electric vehicle energy consumption prediction.

The specific objectives are as follows:

- 1) To develop a comprehensive EV energy prediction framework using vehicle telemetry, battery characteristics, environmental conditions, and road attributes.
- 2) To incorporate physics-informed features derived from vehicle dynamics equations into the machine learning pipeline.
- 3) To evaluate the effectiveness of hybrid physics-informed learning compared with raw-feature and physics-only approaches.
- 4) To design a federated learning architecture that enables distributed model training without sharing raw vehicle data.
- 5) To integrate the trained model with the Simulation of Urban Mobility (SUMO) platform for real-time energy prediction.
- 6) To develop an interactive analytics dashboard for model evaluation and visualization.

#### G. Major Contributions

The major contributions of this work are summarized as follows:

- 1) A physics-informed machine learning framework is proposed for electric vehicle energy consumption prediction by integrating aerodynamic drag, rolling resistance, grade resistance, and inertial power features with data-driven learning.
- 2) A privacy-preserving federated learning architecture is developed using vehicle-based client partitioning across twenty distributed clients.
- 3) A comprehensive ablation study is conducted to evaluate the contribution of raw features, reduced-feature configurations, physics-only features, and hybrid physics-informed features.
- 4) The proposed hybrid model achieves superior performance with a Mean Absolute Error (MAE) of 0.4629 kWh, a Root Mean Square Error (RMSE) of 0.5744 kWh, and an  $R^2$  score of 0.9320.
- 5) The federated implementation achieves comparable predictive performance with an MAE of 0.4844 kWh, demonstrating the feasibility of privacy-preserving distributed learning.
- 6) Real-time deployment is demonstrated through integration with the SUMO traffic simulation platform and an interactive analytics dashboard.

#### H. Paper Organization

The remainder of this paper is organized as follows. Section II presents a comprehensive review of related work on EV energy prediction, physics-informed machine learning, and federated learning. Section III describes the proposed system architecture. Section IV presents the mathematical formulation of the proposed physics-informed features. Section V describes the dataset and preprocessing pipeline. Section VI presents the physics-informed feature engineering process. Section VII describes the federated learning framework. Section VIII presents the experimental setup. Section IX discusses the experimental results and comparative analysis. Section X presents the SUMO-based deployment framework. Section XI describes dashboard analytics and visualization. Section XII discusses study limitations. Section XIII outlines future research directions. Finally, Section XIV concludes the paper and outlines future research directions.

## II. RELATED WORK

### A. Evolution of Electric Vehicle Energy Consumption Prediction

The rapid growth of electric vehicles has transformed transportation systems and created new challenges in energy management, charging infrastructure planning, battery optimization, and intelligent mobility services. Accurate prediction of electric vehicle energy consumption has therefore become a critical research area due to its direct influence on route planning, battery management systems, charging scheduling, fleet operations, and smart grid integration.

Unlike conventional vehicles, whose fuel consumption can often be estimated using relatively simple empirical models, electric vehicle energy consumption depends on a large number of interacting variables. Vehicle speed, acceleration behavior, battery condition, traffic density, road geometry, weather conditions, vehicle mass, tire characteristics, and driving style all

contribute significantly to overall energy usage. The highly nonlinear interactions among these variables make energy prediction a challenging problem and have motivated researchers to develop increasingly sophisticated prediction methodologies.

Over the past decade, research in EV energy prediction has evolved from traditional analytical approaches toward advanced machine learning and artificial intelligence frameworks. Early studies primarily relied on vehicle dynamics equations and physical models to estimate energy demand. Although these methods provided interpretability and consistency with established engineering principles, they often struggled to capture the complexity of real-world driving environments. The availability of large-scale vehicle telemetry datasets subsequently enabled the adoption of data-driven machine learning approaches capable of learning predictive patterns directly from operational data. More recently, researchers have begun exploring hybrid approaches that combine physical knowledge with machine learning techniques, aiming to achieve both predictive accuracy and physical consistency.

The evolution of EV energy prediction methodologies reflects a broader transition occurring across intelligent transportation systems, where purely analytical models are increasingly being complemented by data-driven approaches capable of adapting to complex and dynamic environments.

### *B. Physics-Based Energy Consumption Models*

Physics-based energy consumption models constitute the foundation of electric vehicle energy analysis. These approaches estimate vehicle energy demand by explicitly modeling the physical forces that influence vehicle motion. Such models are derived from classical vehicle dynamics and energy balance principles and provide a direct mathematical relationship between vehicle operation and energy consumption.

The most commonly modeled forces include aerodynamic drag, rolling resistance, grade resistance, and inertial forces. Aerodynamic drag becomes increasingly important at higher vehicle speeds and represents the resistance generated by airflow around the vehicle body. Rolling resistance results from tire deformation and road surface interaction, while grade resistance captures the impact of road inclines and declines. Inertial forces represent the energy required to accelerate or decelerate the vehicle.

Physics-based models have several advantages. Their outputs can be directly interpreted in terms of known engineering principles, making them attractive for battery management systems and energy optimization applications. Furthermore, these models often require relatively small amounts of data compared with machine learning approaches and can be implemented efficiently in embedded systems.

However, the predictive performance of purely physics-based methods depends heavily on accurate parameter estimation. Variables such as drag coefficients, tire conditions, vehicle loading, environmental factors, and battery efficiency may vary significantly under real-world operating conditions. Small inaccuracies in these parameters can result in substantial prediction errors. Moreover, physics-based approaches often struggle to account for driver behavior, traffic interactions, and environmental variability, which play important roles in determining real-world energy consumption.

Consequently, while physics-based models provide valuable interpretability and engineering insight, they are often insufficient when applied independently to large-scale transportation environments characterized by complex and rapidly changing operating conditions.

### *C. Machine Learning-Based Energy Prediction*

The emergence of machine learning has significantly transformed the field of EV energy prediction. Machine learning algorithms are capable of discovering complex nonlinear relationships among variables without requiring explicit mathematical descriptions of the underlying physical processes. This capability has made machine learning one of the most widely adopted approaches for energy prediction in modern intelligent transportation systems.

Early machine learning studies utilized linear regression, decision trees, support vector machines, and nearest-neighbor algorithms to estimate vehicle energy consumption. These methods demonstrated promising results compared with traditional analytical approaches and highlighted the potential of data-driven learning for transportation applications. As computational resources and data availability increased, more sophisticated algorithms were introduced to improve predictive performance.

Ensemble learning methods have become particularly popular in EV energy prediction research. Algorithms such as Random Forests, Gradient Boosting Machines, XGBoost, and LightGBM combine multiple weak learners to construct highly accurate predictive models. These approaches are capable of handling large-scale datasets, nonlinear relationships, and high-dimensional feature spaces while maintaining computational efficiency.

Among ensemble learning techniques, LightGBM has attracted considerable attention due to its ability to process structured tabular data efficiently. The algorithm employs gradient-based optimization strategies that reduce computational complexity while maintaining strong predictive performance. Several studies have demonstrated that LightGBM consistently achieves competitive accuracy across a variety of transportation and energy forecasting applications.

Machine learning approaches provide significant advantages over traditional analytical models because they can learn directly from historical vehicle data and adapt to diverse operating environments. Nevertheless, these models often function as black-box

systems, making it difficult to interpret their internal decision-making processes. In addition, purely data-driven models may exhibit poor generalization when exposed to operating conditions that differ substantially from those observed during training.

These limitations have motivated the development of hybrid methodologies that seek to combine the predictive strength of machine learning with the interpretability of physical models.

#### *D. Deep Learning Approaches*

Deep learning represents a further evolution of data-driven energy prediction methodologies. Unlike traditional machine learning algorithms that rely heavily on manually engineered features, deep learning models automatically learn hierarchical feature representations directly from raw data. This capability enables them to capture complex temporal and spatial relationships present within transportation datasets.

Feedforward neural networks were among the earliest deep learning architectures applied to EV energy prediction. Subsequent developments introduced recurrent neural networks, Long Short-Term Memory networks, and Gated Recurrent Units, which are particularly effective for modeling sequential driving data. These architectures can capture temporal dependencies among vehicle states, battery conditions, and environmental variables, allowing them to model energy consumption patterns over time.

Deep learning models have demonstrated impressive predictive performance when trained on large datasets. Their ability to automatically learn feature representations reduces the need for extensive manual feature engineering and allows them to capture complex interactions that may be difficult to represent analytically.

Despite these advantages, deep learning approaches often require substantial computational resources and large quantities of training data. Training procedures may involve extensive hyperparameter tuning and longer computational times compared with traditional machine learning algorithms. Furthermore, the lack of interpretability associated with many deep learning architectures remains a significant challenge, particularly in transportation systems where transparency and explainability are important for operational decision-making.

As a result, researchers have increasingly explored methods that combine the strengths of physical modeling, machine learning, and deep learning into unified prediction frameworks.

#### *E. Physics-Informed Machine Learning for EV Energy Prediction*

The limitations of purely physics-based models and purely data-driven machine learning approaches have motivated the emergence of Physics-Informed Machine Learning (PIML). This paradigm seeks to integrate established physical principles with artificial intelligence techniques in order to achieve both predictive accuracy and physical consistency. Rather than treating physical modeling and machine learning as independent methodologies, physics-informed learning combines the strengths of both approaches within a unified framework.

The fundamental idea behind physics-informed learning is that machine learning models should not rely exclusively on statistical correlations present within the training data. Instead, they should be guided by domain knowledge derived from scientific principles and engineering theory. In the context of electric vehicle energy prediction, this involves incorporating variables that explicitly represent vehicle dynamics and energy consumption mechanisms.

Several recent studies have demonstrated that incorporating physics-derived features into machine learning pipelines can significantly improve prediction accuracy. Features such as aerodynamic drag power, rolling resistance power, grade resistance power, and inertial power provide physically meaningful representations of vehicle energy demand. Unlike raw telemetry measurements alone, these engineered features directly describe the physical processes responsible for energy consumption.

Physics-informed models also offer advantages in terms of generalization. Since the learning process is partially constrained by physical principles, the resulting models are often less susceptible to overfitting and can perform more reliably when exposed to previously unseen operating conditions. This characteristic is particularly important in transportation systems, where driving environments, weather conditions, and traffic patterns may vary substantially over time.

Another significant advantage of physics-informed learning is improved interpretability. Transportation engineers and energy researchers often require an understanding of why a model produces a particular prediction. By incorporating physically meaningful variables into the learning process, physics-informed frameworks provide greater transparency than conventional black-box machine learning approaches.

Despite these advantages, the application of physics-informed machine learning within electric vehicle energy prediction remains relatively limited. Most existing studies either focus on purely analytical methods or exclusively data-driven approaches. Consequently, there remains considerable opportunity to investigate hybrid frameworks that combine vehicle dynamics-based feature engineering with modern machine learning algorithms in order to improve both accuracy and interpretability.

#### *F. Federated Learning in Intelligent Transportation Systems*

The rapid digitalization of transportation systems has resulted in the generation of unprecedented volumes of data. Modern vehicles continuously collect information related to vehicle operation, battery performance, navigation patterns, environmental conditions, driver behavior, and traffic interactions. Such data provides valuable opportunities for developing advanced predictive models; however, it also raises significant concerns regarding privacy, security, ownership, and regulatory compliance.

Traditional machine learning approaches rely on centralized training, where data from multiple sources is aggregated within a single repository before model development. Although centralized learning often achieves strong predictive performance, it introduces several practical challenges. Large-scale data transfer increases communication costs, exposes sensitive information, and creates potential cybersecurity vulnerabilities. Furthermore, regulatory restrictions may prevent organizations from sharing raw operational data with external entities.

Federated Learning has emerged as a promising solution to these challenges. Federated learning enables multiple distributed clients to collaboratively train machine learning models without exchanging raw data. Instead, each client performs local training using its own dataset and shares only model parameters, gradients, or aggregated predictions. This approach preserves privacy while still allowing knowledge to be extracted from distributed datasets.

The application of federated learning has expanded rapidly across numerous domains, including healthcare, finance, smart manufacturing, and intelligent transportation systems. Within transportation environments, federated learning has been applied to traffic forecasting, vehicle trajectory prediction, autonomous driving systems, and transportation demand modeling. The decentralized nature of transportation data makes federated learning particularly attractive because vehicle information is naturally distributed across individual users, fleet operators, charging stations, and geographic regions.

Several studies have demonstrated that federated learning can achieve predictive performance comparable to centralized learning while significantly reducing privacy risks. However, federated systems introduce new challenges, including communication overhead, client heterogeneity, model synchronization, and computational complexity. Variations in client data distributions can also affect convergence and model stability.

In the context of electric vehicles, federated learning remains an emerging research area. Existing studies have primarily focused on battery health estimation, charging demand forecasting, and driver behavior analysis. Comparatively few investigations have explored federated learning for electric vehicle energy consumption prediction. Furthermore, the integration of federated learning with physics-informed feature engineering remains largely unexplored. This observation highlights a significant research opportunity for developing privacy-preserving energy prediction frameworks that maintain high levels of predictive accuracy.

#### *G. Simulation-Based Evaluation Using SUMO*

Simulation environments have become indispensable tools for transportation research because they enable controlled experimentation under realistic traffic conditions. Evaluating predictive algorithms directly in real-world transportation systems is often expensive, time-consuming, and operationally challenging. Consequently, researchers frequently employ simulation platforms to assess the performance of intelligent transportation technologies before deployment.

Among available simulation environments, the Simulation of Urban Mobility (SUMO) has emerged as one of the most widely used open-source microscopic traffic simulators. SUMO enables detailed modeling of individual vehicle movements, road networks, traffic interactions, traffic control systems, and mobility patterns. The platform supports large-scale transportation simulations and has become a standard tool for intelligent transportation research.

SUMO offers several capabilities that are particularly relevant for electric vehicle studies. Vehicle trajectories, speeds, accelerations, traffic densities, route selections, and environmental interactions can all be simulated with high temporal resolution. These capabilities allow researchers to evaluate energy consumption models under diverse traffic scenarios and operational conditions.

Recent studies have integrated machine learning techniques with SUMO in order to investigate traffic prediction, congestion management, route optimization, and autonomous driving systems. For electric vehicle applications, SUMO has been used to simulate charging behavior, battery usage patterns, energy consumption, and charging infrastructure deployment strategies. Such simulations provide valuable insights into how predictive models behave under realistic transportation conditions.

Despite the widespread use of SUMO, relatively few studies have combined traffic simulation with physics-informed machine learning and federated learning frameworks. Most existing simulation-based investigations focus either on transportation optimization or machine learning evaluation independently. The integration of advanced predictive models with real-time transportation simulation therefore represents an important research direction that can bridge the gap between algorithm development and practical deployment.

Furthermore, simulation-based evaluation enables researchers to examine system behavior under conditions that may be difficult or costly to reproduce in real-world environments. As intelligent transportation systems continue to evolve, simulation platforms such as SUMO will remain essential tools for validating next-generation energy prediction frameworks.

#### *H. Visualization and Decision-Support Dashboards*

The increasing complexity of transportation systems has created a growing demand for effective visualization and decision-support tools. Predictive models are most valuable when their outputs can be interpreted and utilized by transportation planners, fleet operators, charging infrastructure managers, and policymakers. Consequently, visualization dashboards have become an important component of modern transportation analytics platforms.

Interactive dashboards provide users with real-time access to predictive insights, performance metrics, operational statistics, and system-level indicators. In the context of electric vehicles, dashboards can be used to monitor energy consumption, battery status, charging behavior, vehicle efficiency, route performance, and prediction accuracy. Such capabilities support informed decision-making and improve operational transparency.

Recent advances in web-based analytics frameworks have enabled the development of highly interactive visualization platforms that integrate machine learning outputs with transportation data streams. These systems facilitate model monitoring, anomaly detection, performance evaluation, and user engagement. Visual representations such as distribution plots, feature importance analyses, residual error distributions, correlation heatmaps, and real-time prediction displays help stakeholders understand model behavior and assess system performance.

The integration of predictive analytics with visualization tools is particularly important for intelligent transportation systems because it enables complex machine learning outputs to be communicated effectively to both technical and non-technical audiences. Consequently, modern transportation research increasingly emphasizes not only prediction accuracy but also interpretability, transparency, and usability.

### *I. Research Gap Analysis*

The literature reviewed in this section demonstrates significant progress in electric vehicle energy consumption prediction, machine learning, federated learning, and transportation simulation. Nevertheless, several important challenges remain unresolved.

First, existing physics-based approaches provide strong interpretability but often struggle to capture the complex nonlinear relationships present in real-world transportation systems. Conversely, purely data-driven machine learning approaches achieve high predictive performance but frequently lack physical consistency and interpretability. This creates a need for hybrid frameworks capable of combining physical knowledge with machine learning techniques.

Second, despite the increasing popularity of federated learning, relatively few studies have investigated its application to electric vehicle energy consumption prediction. Most existing work focuses on centralized learning architectures that require extensive data sharing and raise privacy concerns. The development of privacy-preserving prediction frameworks therefore remains an important research priority.

Third, while transportation simulation platforms such as SUMO are widely used for mobility analysis, limited research has integrated physics-informed machine learning models, federated learning architectures, and real-time traffic simulation within a unified framework. Consequently, the practical deployment readiness of many existing approaches remains uncertain.

Finally, many existing studies focus exclusively on predictive accuracy while overlooking interpretability, scalability, privacy preservation, and real-time decision support. Future transportation systems require integrated solutions capable of addressing all of these requirements simultaneously.

To address these research gaps, this study proposes a Physics-Informed Federated Learning Framework for Electric Vehicle Energy Consumption Prediction with Real-Time Traffic Simulation. The proposed framework integrates vehicle dynamics-based feature engineering, LightGBM-based prediction, federated learning, SUMO simulation, and dashboard analytics within a unified architecture. By combining these components, the framework aims to achieve high predictive accuracy while maintaining physical interpretability, privacy preservation, scalability, and practical deployment readiness.

## III. SYSTEM ARCHITECTURE

### *A. Overview*

The proposed system architecture is designed to provide an end-to-end framework for accurate, interpretable, privacy-preserving, and deployment-ready electric vehicle energy consumption prediction. The architecture integrates physics-informed feature engineering, machine learning-based prediction, federated learning, real-time traffic simulation, and interactive analytics within a unified framework.

The complete architecture consists of six major layers:

- 1) Data Acquisition Layer
- 2) Data Preprocessing Layer
- 3) Physics-Informed Feature Engineering Layer
- 4) Machine Learning and Federated Learning Layer
- 5) Real-Time Simulation Layer
- 6) Visualization and Analytics Layer

The overall workflow of the proposed framework is illustrated in Fig. 1.

Overall System Architecture of the Proposed Framework

Fig. 1. Overall system architecture of the proposed framework.

### B. Data Acquisition Layer

The first layer of the proposed architecture consists of the electric vehicle operational dataset. The dataset contains 5,000 vehicle records collected from simulated electric vehicle operating conditions. Each record contains information related to vehicle dynamics, battery characteristics, environmental conditions, road attributes, and travel behavior.

The dataset includes variables such as vehicle speed, acceleration, battery state, battery voltage, battery temperature, road slope, weather conditions, humidity, wind speed, tire pressure, vehicle weight, and travelled distance. The target variable is vehicle energy consumption measured in kilowatt-hours (kWh).

This layer serves as the primary source of information for all subsequent processing and modeling stages.

### C. Data Preprocessing Layer

Raw transportation datasets frequently contain inconsistencies, redundant information, and noisy observations that can negatively affect model performance. Therefore, a preprocessing stage is incorporated before model development.

The preprocessing pipeline performs data loading, cleaning, validation, and feature selection. Non-predictive identifiers such as Vehicle\_ID and Timestamp are excluded from the machine learning process because they do not contribute directly to energy consumption prediction.

The resulting dataset provides a clean and structured foundation for subsequent feature engineering and model training procedures.

### D. Physics-Informed Feature Engineering Layer

One of the most important components of the proposed framework is the incorporation of physics-informed features derived from vehicle dynamics principles. Instead of relying exclusively on raw telemetry measurements, additional physically meaningful variables are generated using established energy consumption equations.

The following physics-based features are computed:

- Aerodynamic Drag Power ( $P_{\text{drag}}$ )
- Rolling Resistance Power ( $P_{\text{roll}}$ )
- Grade Resistance Power ( $P_{\text{grade}}$ )
- Inertial Power ( $P_{\text{inertia}}$ )
- Total Vehicle Power ( $P_{\text{total}}$ )

These features provide explicit representations of the physical mechanisms responsible for vehicle energy consumption. The resulting hybrid feature space combines both measured vehicle telemetry and physics-derived variables, enabling the machine learning model to learn relationships that are both statistically meaningful and physically interpretable.

### E. Machine Learning Layer

The machine learning layer is responsible for learning the relationship between vehicle operating conditions and energy consumption. A Light Gradient Boosting Machine (LightGBM) regressor is selected as the primary prediction algorithm due to its strong performance on structured tabular datasets, computational efficiency, and ability to model nonlinear relationships.

The model is trained using the hybrid feature dataset consisting of both raw variables and physics-informed features. Several model configurations are evaluated, including:

- Raw Feature Model
- Reduced Feature Model
- Physics-Only Model
- Hybrid Physics-Informed Model

This comparative evaluation enables assessment of the contribution of different feature groups toward predictive performance.

### F. Federated Learning Layer

To address privacy concerns associated with centralized data collection, the proposed architecture incorporates a federated learning framework. The dataset is partitioned according to vehicle identifiers and distributed across twenty federated clients. Each client performs local model training using only its own subset of vehicle records.

Rather than sharing raw data, clients contribute local predictive outputs that are combined through weighted aggregation. The aggregation weight of each client is proportional to the number of training samples available within that client. This approach preserves data privacy while still enabling collaborative model development.

The federated layer enables the framework to simulate realistic large-scale deployment scenarios where vehicle data remains distributed across multiple sources.



### G. SUMO-Based Simulation Layer

The trained prediction model is integrated with the Simulation of Urban Mobility (SUMO) platform in order to evaluate real-time deployment capabilities. SUMO provides a microscopic traffic simulation environment capable of modeling vehicle movement, traffic interactions, route selection, and transportation dynamics.

During simulation, real-time vehicle information such as speed and acceleration is continuously collected and supplied to the prediction model. The model estimates instantaneous energy consumption for each active vehicle within the simulation environment.

This integration enables assessment of model performance under realistic traffic conditions and demonstrates practical deployment feasibility.

### H. Dashboard Analytics Layer

The final layer of the architecture consists of an interactive dashboard developed for visualization and decision support. The dashboard provides:

- Vehicle-level energy analytics
- Prediction performance visualization
- Error distribution analysis
- Actual versus predicted comparisons
- Model comparison statistics
- Fleet-level energy monitoring

The visualization layer transforms model outputs into intuitive graphical representations that support both technical and non-technical stakeholders. This component improves transparency, interpretability, and usability of the proposed framework.

### I. Architectural Advantages

The proposed architecture offers several advantages over conventional EV energy prediction systems. First, the integration of physics-informed features improves model interpretability and predictive performance. Second, federated learning enables privacy-preserving model development without requiring centralized data collection. Third, SUMO integration provides a realistic deployment environment for evaluating real-time performance. Finally, the dashboard analytics layer enhances accessibility and supports informed decision-making through interactive visualization.

Collectively, these components form a comprehensive framework for intelligent, scalable, and privacy-aware electric vehicle energy consumption prediction.

## IV. MATHEMATICAL FORMULATION

### A. Overview

The proposed Physics-Informed Federated Learning Framework integrates classical vehicle dynamics principles with machine learning techniques to predict electric vehicle energy consumption. Rather than relying solely on raw telemetry measurements, physically meaningful variables are derived from established vehicle energy equations and incorporated into the prediction framework.

The mathematical formulation presented in this section describes the computation of the physics-informed features used during model training. These features represent the major sources of energy expenditure experienced by an electric vehicle during operation and form the foundation of the proposed hybrid prediction architecture.

The total power demand of a moving electric vehicle can be represented as the combined effect of aerodynamic drag, rolling resistance, road grade resistance, and inertial forces. Accordingly, the total vehicle power is expressed as:

$$P_{\text{total}} = P_{\text{drag}} + P_{\text{roll}} + P_{\text{grade}} + P_{\text{inertia}}, \quad (1)$$

where  $P_{\text{drag}}$  represents aerodynamic drag power,  $P_{\text{roll}}$  represents rolling resistance power,  $P_{\text{grade}}$  represents grade resistance power, and  $P_{\text{inertia}}$  represents inertial power. These quantities are computed individually and subsequently combined to generate the physics-informed feature set.

### B. Vehicle Speed Conversion

The dataset records vehicle speed in kilometers per hour (km/h). Since vehicle dynamics equations require velocity in meters per second (m/s), a conversion is performed prior to feature computation.

The speed conversion equation is given by:

$$v = \frac{V_{\text{km/h}}}{3.6}, \quad (2)$$

where  $v$  is the vehicle speed in meters per second (m/s), and  $V_{\text{km/h}}$  is the vehicle speed in kilometers per hour (km/h). This converted speed variable serves as the basis for all subsequent power calculations.

### C. Road Slope Representation

Road gradient information is represented as a percentage value within the dataset. To facilitate mathematical computation, the slope percentage is converted into a normalized slope ratio according to:

$$S = \frac{\text{Slope}\%}{100}, \quad (3)$$

where  $S$  is the normalized road slope ratio, and  $\text{Slope}\%$  is the road gradient expressed as a percentage. The slope ratio quantifies the elevation change experienced by the vehicle and directly influences gravitational energy requirements.

### D. Aerodynamic Drag Power

Aerodynamic drag is one of the dominant sources of energy consumption at medium and high vehicle speeds. As the vehicle moves through air, it experiences a resistive force proportional to the square of velocity. Consequently, the required aerodynamic power increases approximately with the cube of velocity.

The aerodynamic drag power is calculated as:

$$P_{\text{drag}} = \frac{1}{2} \rho C_d A v^3, \quad (4)$$

where  $\rho$  is the air density ( $1.225 \text{ kg/m}^3$ ),  $C_d$  is the aerodynamic drag coefficient (0.29),  $A$  is the frontal area of the vehicle ( $2.2 \text{ m}^2$ ), and  $v$  is the vehicle speed (m/s). This equation estimates the power required to overcome air resistance during vehicle motion.

The cubic relationship between velocity and aerodynamic power highlights the substantial increase in energy consumption observed at higher vehicle speeds.

### E. Rolling Resistance Power

Rolling resistance arises from tire deformation and frictional interaction between the tires and road surface. Although smaller than aerodynamic drag at high speeds, rolling resistance contributes continuously throughout vehicle operation.

The rolling resistance power is computed as:

$$P_{\text{roll}} = mg C_r v, \quad (5)$$

where  $m$  is the vehicle mass (kg),  $g$  is the gravitational acceleration ( $9.81 \text{ m/s}^2$ ),  $C_r$  is the rolling resistance coefficient (0.01), and  $v$  is the vehicle speed (m/s). This formulation quantifies the power required to overcome tire-road interaction losses during movement.

Rolling resistance is particularly significant in urban driving conditions where average vehicle speeds remain relatively low.

### F. Grade Resistance Power

Road inclination significantly affects vehicle energy consumption. When a vehicle travels uphill, additional energy is required to overcome gravitational forces. Conversely, downhill travel may reduce energy demand and facilitate regenerative braking.

The grade resistance power is calculated as:

$$P_{\text{grade}} = mg S v, \quad (6)$$

where  $m$  is the vehicle mass (kg),  $g$  is the gravitational acceleration,  $S$  is the normalized slope ratio, and  $v$  is the vehicle speed (m/s). This equation estimates the additional power required due to changes in road elevation.

Grade resistance becomes particularly important in mountainous and hilly environments where elevation variations significantly influence energy consumption.

### G. Inertial Power

Vehicle acceleration and deceleration contribute substantially to energy consumption, particularly in stop-and-go urban traffic conditions.

The inertial power component is computed as:

$$P_{\text{inertia}} = m a v, \quad (7)$$

where  $m$  is the vehicle mass (kg),  $a$  is the vehicle acceleration ( $\text{m/s}^2$ ), and  $v$  is the vehicle speed (m/s). This equation represents the power required to change the kinetic energy state of the vehicle.

Positive acceleration values increase energy demand, while negative acceleration values may correspond to regenerative braking opportunities. Consequently, inertial power serves as a critical indicator of driver behavior and traffic dynamics.

### H. Total Vehicle Power

The individual power components are combined to estimate the total instantaneous vehicle power demand.

The total power equation is expressed as:

$$P_{\text{total}} = P_{\text{drag}} + P_{\text{roll}} + P_{\text{grade}} + P_{\text{inertia}}. \quad (8)$$

Substituting the individual formulations yields:

$$P_{\text{total}} = \frac{1}{2}\rho C_d A v^3 + mg C_r v + mg S v + m a v. \quad (9)$$

This equation forms the foundation of the proposed physics-informed feature engineering framework. The resulting value captures the combined influence of vehicle dynamics, environmental conditions, road characteristics, and driving behavior on energy consumption.

### I. Physics-Informed Feature Vector

The generated physics features are incorporated into the machine learning pipeline as additional explanatory variables.

The final physics feature vector is defined as:

$$F_{\text{physics}} = \{P_{\text{drag}}, P_{\text{roll}}, P_{\text{grade}}, P_{\text{inertia}}, P_{\text{total}}\}. \quad (10)$$

These variables are appended to the original dataset to form a hybrid feature representation. The final machine learning input vector is therefore expressed as:

$$X_{\text{hybrid}} = X_{\text{raw}} \cup F_{\text{physics}}, \quad (11)$$

where  $X_{\text{raw}}$  represents the original vehicle telemetry and environmental variables, and  $F_{\text{physics}}$  represents the physics-informed features. This hybrid representation enables the learning algorithm to simultaneously exploit measured operational data and physically meaningful energy-related information.

### J. Federated Prediction Aggregation

The federated learning framework consists of  $N$  distributed clients. Each client independently trains a local model using its assigned vehicle subset and generates local predictions.

The global federated prediction is computed using weighted aggregation:

$$\hat{y} = \sum_{i=1}^N w_i \hat{y}_i, \quad (12)$$

where  $\hat{y}_i$  is the prediction generated by client  $i$ , and  $w_i$  is the aggregation weight assigned to client  $i$ .

The client weight is calculated according to:

$$w_i = \frac{|D_i|}{\sum_{j=1}^N |D_j|}, \quad (13)$$

where  $|D_i|$  represents the number of samples within client  $i$ . This weighting strategy ensures that larger clients contribute proportionally more to the final prediction while preserving data locality and privacy.

### K. Energy Consumption Prediction Objective

The objective of the proposed framework is to learn a nonlinear mapping:

$$f(X_{\text{hybrid}}) = \hat{E}, \quad (14)$$

where  $X_{\text{hybrid}}$  is the hybrid feature vector, and  $\hat{E}$  is the predicted energy consumption.

The prediction model seeks to minimize the difference between predicted energy consumption and actual energy consumption values observed within the dataset. The resulting framework combines physical interpretability, machine learning accuracy, and federated privacy preservation within a unified mathematical formulation.

TABLE I  
DESCRIPTION OF DATASET FEATURES

| Feature Name           | Description                                                |
|------------------------|------------------------------------------------------------|
| Vehicle_ID             | Unique identifier assigned to each vehicle                 |
| Timestamp              | Temporal information associated with each observation      |
| Speed_kmh              | Vehicle speed measured in kilometers per hour              |
| Acceleration_ms2       | Vehicle acceleration measured in meters per second squared |
| Battery_State_%        | Battery state of charge percentage                         |
| Battery_Voltage_V      | Battery voltage level in volts                             |
| Battery_Temperature_C  | Battery temperature in degrees Celsius                     |
| Driving_Mode           | Vehicle driving mode indicator                             |
| Road_Type              | Classification of road segment                             |
| Traffic_Condition      | Traffic density or congestion level                        |
| Slope_%                | Road gradient percentage                                   |
| Weather_Condition      | Weather category associated with the trip                  |
| Temperature_C          | Ambient temperature                                        |
| Humidity_%             | Relative humidity percentage                               |
| Wind_Speed_ms          | Wind speed measured in meters per second                   |
| Tire_Pressure_psi      | Tire pressure in pounds per square inch                    |
| Vehicle_Weight_kg      | Vehicle mass measured in kilograms                         |
| Distance_Travelled_km  | Travel distance measured in kilometers                     |
| Energy_Consumption_kWh | Energy consumption target variable                         |

## V. DATASET DESCRIPTION AND DATA PREPROCESSING

### A. Dataset Overview

The performance of any machine learning framework is highly dependent on the quality and representativeness of the underlying dataset. In this study, an electric vehicle operational dataset consisting of 5,000 records was utilized to develop and evaluate the proposed Physics-Informed Federated Learning Framework for energy consumption prediction.

The dataset was designed to capture the multidimensional factors that influence electric vehicle energy consumption. These factors include vehicle dynamics, battery characteristics, environmental conditions, road attributes, and trip-related information. The target variable of the dataset is the vehicle energy consumption measured in kilowatt-hours (kWh).

The dataset contains operational information collected under diverse driving scenarios in order to ensure sufficient variability for machine learning model training and evaluation. By incorporating multiple sources of energy-related information, the dataset provides a comprehensive representation of real-world electric vehicle operating conditions.

A total of eighteen predictor variables and one target variable were considered during model development. The predictor variables describe vehicle motion, battery status, weather conditions, road geometry, and vehicle-specific characteristics, all of which contribute directly or indirectly to energy consumption.

The target variable, *Energy\_Consumption\_kWh*, represents the total energy consumed by the vehicle during a particular operational interval and serves as the prediction objective of the proposed framework.

### B. Feature Description

The dataset consists of nineteen columns, including eighteen explanatory variables and one target variable. Table I summarizes the complete feature set used in this study.

The selected variables provide a balanced representation of vehicle dynamics, battery characteristics, environmental factors, and operational conditions. Such diversity enables the prediction model to capture the complex nonlinear relationships governing electric vehicle energy consumption.

### C. Data Cleaning and Preparation

Raw transportation datasets often contain inconsistencies, missing values, duplicate observations, and noisy measurements that can negatively affect machine learning performance. Consequently, a data preprocessing stage was incorporated prior to model training.

The preprocessing pipeline begins by loading the dataset into the analysis environment and validating its structural integrity. During this stage, irrelevant records, inconsistencies, and potential anomalies are identified and corrected. The preprocessing procedure ensures that all variables are represented in a machine-readable format suitable for subsequent feature engineering and model development.

Particular attention was given to maintaining consistency among numerical variables associated with vehicle dynamics, battery parameters, environmental conditions, and trip characteristics. This step is essential because machine learning algorithms are highly sensitive to data quality and may produce unreliable results when trained on noisy or inconsistent datasets.

Following the cleaning stage, the dataset was transformed into a structured format that could be directly utilized by the machine learning pipeline. This standardized representation served as the foundation for the subsequent physics-informed feature engineering process.

#### D. Feature Selection

Not all variables contained within the dataset contribute directly to the prediction process. Certain attributes primarily serve identification or temporal tracking purposes rather than providing predictive information.

Specifically, the variables `Vehicle_ID` and `Timestamp` were excluded from model training. `Vehicle_ID` functions as a unique identifier and does not represent an energy-related characteristic. Similarly, `Timestamp` serves as a temporal reference but does not directly influence energy consumption within the prediction framework.

Removing these variables helps reduce unnecessary complexity and prevents the model from learning patterns that are unrelated to vehicle energy dynamics. Consequently, the final machine learning feature set consists of operational, environmental, battery, and vehicle-related variables that exhibit direct relationships with energy consumption.

#### E. Physics-Informed Feature Generation

A distinguishing characteristic of the proposed framework is the integration of physics-informed features into the machine learning pipeline. Instead of relying exclusively on raw telemetry measurements, additional variables derived from vehicle dynamics principles were generated.

Five physics-based features were computed:

- Aerodynamic Drag Power ( $P_{\text{drag}}$ )
- Rolling Resistance Power ( $P_{\text{roll}}$ )
- Grade Resistance Power ( $P_{\text{grade}}$ )
- Inertial Power ( $P_{\text{inertia}}$ )
- Total Vehicle Power ( $P_{\text{total}}$ )

These variables represent the major physical mechanisms responsible for electric vehicle energy expenditure. By incorporating these features, the resulting dataset combines measured operational variables with physically meaningful energy indicators.

The inclusion of physics-informed features enables the prediction model to learn relationships that are both statistically significant and physically interpretable.

#### F. Hybrid Feature Dataset

Following the feature engineering stage, the original dataset and the generated physics-informed variables were merged to form a hybrid feature representation.

The hybrid dataset consists of:

- 1) Raw operational features.
- 2) Battery-related variables.
- 3) Environmental variables.
- 4) Road-condition variables.
- 5) Physics-informed energy features.

This combined representation provides a richer description of vehicle behavior than either raw features or physics-based features alone. The hybrid dataset serves as the primary input for the centralized LightGBM model and the federated learning framework developed in this study.

#### G. Training and Testing Partition

To evaluate model performance objectively, the dataset was divided into training and testing subsets using an 80:20 partitioning strategy.

The training dataset contains:

$$4000 \text{ samples}, \quad (15)$$

while the testing dataset contains:

$$1000 \text{ samples}. \quad (16)$$

A random seed value of 42 was used during partitioning to ensure reproducibility of experimental results. The training subset was utilized for model development and parameter learning, whereas the testing subset remained unseen during training and was used exclusively for performance evaluation.

This separation ensures that reported performance metrics accurately reflect the generalization capability of the proposed framework.

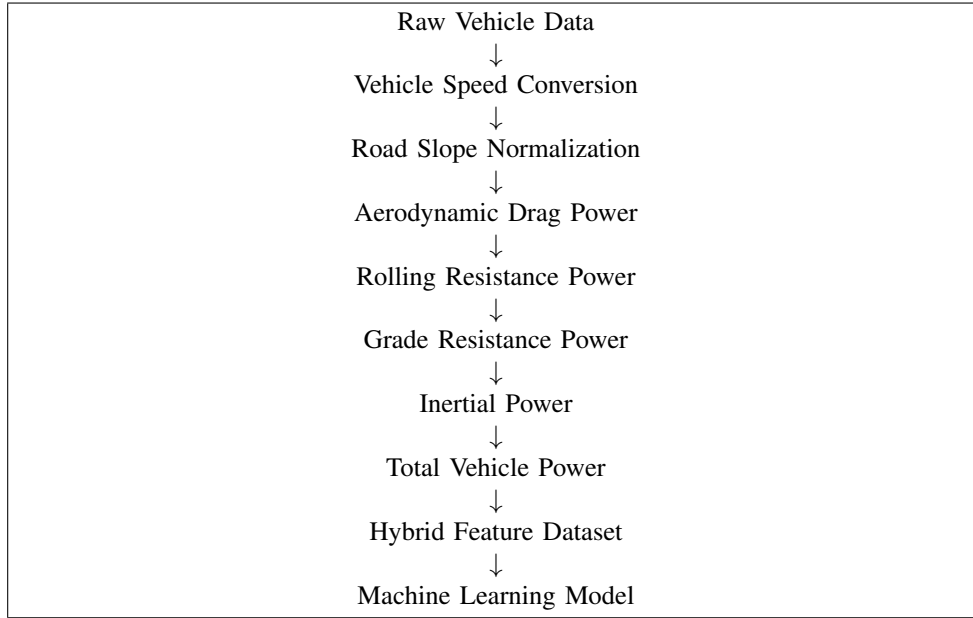


Fig. 2. Physics-informed feature engineering pipeline.

#### H. Summary

The dataset and preprocessing pipeline establish the foundation of the proposed energy prediction framework. A dataset containing 5,000 electric vehicle operational records was cleaned, validated, and transformed into a structured machine learning format. Relevant predictive variables were retained, while non-informative identifiers were excluded. Physics-informed features derived from vehicle dynamics principles were subsequently generated and combined with raw operational variables to create a hybrid feature dataset. Finally, the data was partitioned into training and testing subsets to facilitate robust model evaluation and comparative analysis.

## VI. PHYSICS-INFORMED FEATURE ENGINEERING

### A. Introduction

One of the primary limitations of conventional machine learning models is their dependence on statistical relationships extracted directly from data. Although such models often achieve high predictive accuracy, they frequently lack physical interpretability and may generate unrealistic predictions when exposed to operating conditions that differ from the training data. In the context of electric vehicle energy consumption prediction, this limitation becomes particularly important because vehicle energy demand is fundamentally governed by well-established physical laws.

To address this challenge, the proposed framework incorporates a Physics-Informed Feature Engineering (PIFE) module. The objective of this module is to augment the original dataset with physically meaningful variables derived from vehicle dynamics principles. Rather than relying exclusively on raw telemetry measurements, the framework explicitly models the major physical forces that contribute to vehicle energy expenditure.

The resulting physics-informed features provide additional information regarding the underlying mechanisms responsible for energy consumption and enable the machine learning model to learn relationships that are both statistically significant and physically meaningful.

The complete feature engineering workflow is illustrated in Fig. 2.

### B. Motivation for Physics-Informed Learning

Electric vehicle energy consumption is determined by the interaction of multiple physical forces acting on the vehicle during operation. While machine learning algorithms can learn correlations between measured variables and energy demand, they do not inherently understand the physical principles governing vehicle motion.

For example, vehicle speed influences aerodynamic drag, acceleration affects inertial energy requirements, and road gradients alter gravitational energy demand. These relationships are dictated by fundamental laws of mechanics and remain valid regardless of the specific dataset being analyzed.

Incorporating physics-informed features provides several advantages:

- Improved model interpretability.
- Enhanced generalization capability.

- Reduced dependence on purely statistical correlations.
- Better representation of real-world vehicle dynamics.
- Increased robustness under varying operating conditions.

Consequently, the proposed framework integrates vehicle dynamics equations directly into the feature engineering process.

### C. Vehicle Speed Transformation

The dataset records vehicle velocity in kilometers per hour (km/h). However, vehicle dynamics equations require speed to be represented in meters per second (m/s). Therefore, vehicle speed is transformed according to:

$$v = \frac{\text{Speed}_{\text{km/h}}}{3.6}, \quad (17)$$

where  $v$  denotes vehicle velocity in meters per second, and  $\text{Speed}_{\text{km/h}}$  represents the measured vehicle speed. This transformation ensures consistency with the International System of Units (SI) and facilitates subsequent power calculations.

The converted speed variable serves as the foundation for all physics-based feature computations.

### D. Road Slope Normalization

Road gradient information is represented within the dataset as a percentage value. Since power calculations require a normalized representation of slope, the percentage value is converted into a slope ratio.

The normalized slope is calculated as:

$$\text{Slope}_{\text{ratio}} = \frac{\text{Slope}_{\%}}{100}, \quad (18)$$

where  $\text{Slope}_{\%}$  represents the road gradient percentage, and  $\text{Slope}_{\text{ratio}}$  represents the normalized slope value. This transformation enables the direct incorporation of road elevation effects into vehicle power calculations.

### E. Aerodynamic Drag Power

Aerodynamic drag represents the resistive force generated as a vehicle moves through air. The magnitude of this force increases rapidly with speed and becomes one of the dominant contributors to energy consumption during highway driving.

The aerodynamic drag power is computed using:

$$P_{\text{drag}} = \frac{1}{2} \rho C_d A v^3, \quad (19)$$

where  $\rho = 1.225 \text{ kg/m}^3$  represents air density,  $C_d = 0.29$  denotes the aerodynamic drag coefficient,  $A = 2.2 \text{ m}^2$  represents the frontal area of the vehicle, and  $v$  is the vehicle speed.

The cubic dependence on velocity highlights the substantial increase in energy demand at higher speeds. Consequently, aerodynamic drag power serves as an important indicator of highway energy consumption behavior.

### F. Rolling Resistance Power

Rolling resistance arises from tire deformation and frictional interaction between the tire and road surface. Unlike aerodynamic drag, rolling resistance affects the vehicle throughout its operation regardless of driving speed.

The rolling resistance power is calculated as:

$$P_{\text{roll}} = mg C_r v, \quad (20)$$

where  $m$  denotes vehicle mass,  $g = 9.81 \text{ m/s}^2$  represents gravitational acceleration,  $C_r = 0.01$  denotes the rolling resistance coefficient, and  $v$  represents vehicle speed.

Rolling resistance contributes continuously to energy expenditure and becomes particularly significant during low-speed urban driving conditions.

### G. Grade Resistance Power

Road elevation changes have a direct impact on vehicle energy consumption. Vehicles traveling uphill must overcome gravitational forces, resulting in increased energy demand.

The grade resistance power is calculated using:

$$P_{\text{grade}} = mg \text{Slope}_{\text{ratio}} v, \quad (21)$$

where  $m$  represents vehicle mass,  $g$  represents gravitational acceleration,  $\text{Slope}_{\text{ratio}}$  denotes normalized road gradient, and  $v$  represents vehicle velocity.

This feature captures the additional energy required to overcome road inclines and provides valuable information regarding terrain-related energy consumption. In regions characterized by hilly or mountainous terrain, grade resistance often becomes one of the most influential determinants of vehicle energy demand.

### H. Inertial Power

Vehicle acceleration and deceleration significantly influence energy consumption, particularly in urban environments characterized by frequent stop-and-go traffic conditions.

The inertial power component is computed as:

$$P_{\text{inertia}} = mav, \quad (22)$$

where  $m$  denotes vehicle mass,  $a$  represents vehicle acceleration, and  $v$  denotes vehicle speed.

Positive acceleration values increase energy demand because additional power is required to increase vehicle kinetic energy. Conversely, negative acceleration values may correspond to regenerative braking events capable of recovering energy. Consequently, inertial power provides a useful representation of driver behavior and traffic dynamics.

### I. Total Vehicle Power

While individual power components provide insights into specific energy consumption mechanisms, a comprehensive representation of vehicle power demand requires combining all major forces acting on the vehicle.

The total vehicle power is therefore computed as:

$$P_{\text{total}} = P_{\text{drag}} + P_{\text{roll}} + P_{\text{grade}} + P_{\text{inertia}}. \quad (23)$$

Substituting the individual formulations yields:

$$P_{\text{total}} = \frac{1}{2}\rho C_d A v^3 + mg C_r v + mg \text{Slope}_{\text{ratio}} v + mav. \quad (24)$$

This feature provides a holistic estimate of the instantaneous power demand experienced by the vehicle. Among all generated physics-informed variables, total vehicle power is expected to exhibit one of the strongest relationships with actual energy consumption.

### J. Hybrid Feature Construction

After computing the physics-based variables, the generated features are combined with the original dataset to create a hybrid feature representation.

The final physics-informed feature set consists of:

$$F_{\text{physics}} = \{P_{\text{drag}}, P_{\text{roll}}, P_{\text{grade}}, P_{\text{inertia}}, P_{\text{total}}\}. \quad (25)$$

The complete hybrid input vector is defined as:

$$X_{\text{hybrid}} = X_{\text{raw}} \cup F_{\text{physics}}, \quad (26)$$

where  $X_{\text{raw}}$  represents the original vehicle telemetry and environmental variables, and  $F_{\text{physics}}$  represents the engineered physics-based variables.

This hybrid representation enables the prediction model to simultaneously exploit measured operational information and physically meaningful energy indicators.

### K. Impact of Physics-Informed Features

To evaluate the effectiveness of physics-informed feature engineering, multiple model configurations were investigated. The experimental results demonstrated that models utilizing only raw features or only physics-based features exhibited lower predictive performance compared with the hybrid approach. By combining both sources of information, the model was able to capture complex nonlinear relationships while maintaining physical consistency.

The hybrid model achieved the lowest prediction error among all evaluated configurations, confirming the effectiveness of integrating physics-informed variables within the machine learning pipeline. These findings indicate that physics-informed feature engineering constitutes a critical component of the proposed framework and contributes significantly to its predictive capability.

### L. Summary

This section introduced the Physics-Informed Feature Engineering module of the proposed framework. Five physically meaningful variables were derived from established vehicle dynamics principles, including aerodynamic drag power, rolling resistance power, grade resistance power, inertial power, and total vehicle power. These variables were combined with the original operational dataset to create a hybrid feature representation that captures both statistical and physical characteristics of vehicle energy consumption. The resulting feature set forms the foundation of the subsequent machine learning and federated learning models developed in this study.



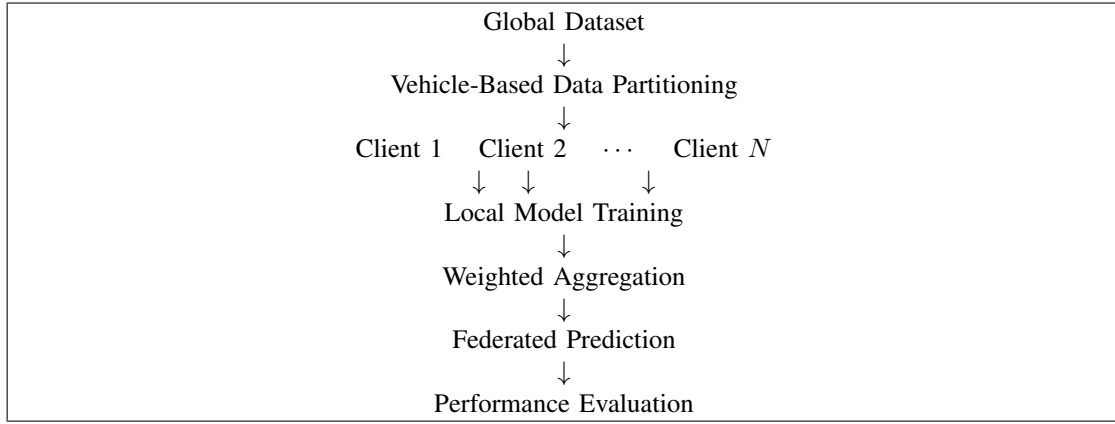


Fig. 3. Federated learning architecture.

## VII. FEDERATED LEARNING FRAMEWORK

### A. Introduction

The rapid growth of connected electric vehicles has resulted in the generation of massive volumes of operational data. Modern vehicles continuously collect information related to battery performance, driving behavior, environmental conditions, navigation patterns, and vehicle dynamics. While these data sources provide valuable opportunities for developing advanced predictive models, they also introduce significant concerns regarding privacy, security, data ownership, and regulatory compliance.

Traditional machine learning approaches typically rely on centralized training, where data from multiple sources is aggregated into a single repository prior to model development. Although centralized learning often achieves strong predictive performance, it requires extensive data sharing and creates potential privacy risks. In large-scale electric vehicle ecosystems, vehicle manufacturers, fleet operators, charging providers, and transportation agencies may be unwilling or unable to share raw operational data due to legal, commercial, or privacy-related constraints.

To address these challenges, the proposed framework incorporates Federated Learning (FL), a distributed machine learning paradigm that enables collaborative model development without requiring the exchange of raw data. Instead of transferring vehicle records to a central server, local models are trained independently on distributed datasets, and their outputs are aggregated to generate a global prediction model.

The federated learning component of the proposed framework is designed to preserve privacy while maintaining predictive performance comparable to centralized learning approaches.

### B. Federated Learning Architecture

The federated learning architecture consists of multiple distributed clients and a global aggregation mechanism. Each client represents a subset of electric vehicles and maintains its local dataset independently. Model training occurs locally within each client environment, ensuring that raw vehicle data never leaves the client.

The overall federated learning workflow is illustrated in Fig. 3.

Unlike centralized learning, vehicle data remains distributed across clients throughout the training process. This design significantly reduces privacy risks while preserving the benefits of collaborative learning.

### C. Vehicle-Based Client Partitioning

A key component of the proposed federated framework is the partitioning strategy used to create federated clients. Rather than randomly splitting individual records, the dataset is partitioned according to vehicle identifiers. This approach ensures that all observations associated with a particular vehicle remain within the same client.

Let

$$V = \{v_1, v_2, v_3, \dots, v_n\} \quad (27)$$

represent the set of unique vehicle identifiers contained within the dataset.

The vehicle set is randomly shuffled and divided into multiple client groups:

$$C = \{C_1, C_2, C_3, \dots, C_N\}, \quad (28)$$

where  $C_i$  represents the  $i^{\text{th}}$  federated client, and  $N$  denotes the total number of clients. In the proposed framework:

$$N = 20. \quad (29)$$

TABLE II  
LOCAL MODEL CONFIGURATION

| Parameter          | Value    |
|--------------------|----------|
| Learning Algorithm | LightGBM |
| Number of Trees    | 200      |
| Learning Rate      | 0.05     |
| Maximum Tree Depth | 6        |

Thus, twenty federated clients are created from the available vehicle population. Each client receives all records corresponding to its assigned vehicle subset. This partitioning strategy more accurately reflects real-world deployment scenarios in which vehicle data naturally remains associated with individual users, fleet operators, or geographical regions.

#### D. Local Model Training

After client creation, each federated client independently trains a local prediction model using only its assigned data. The local training process follows the same machine learning pipeline employed by the centralized framework.

For each client, the local training process consists of the following steps:

- 1) Raw features are loaded.
- 2) Physics-informed features are generated.
- 3) Feature selection is performed.
- 4) Local model training is executed.

The prediction model employed at each client is a Light Gradient Boosting Machine (LightGBM) regressor. The local model configuration is summarized in Table II.

The local training process can be represented as:

$$M_i = \text{Train}(D_i), \quad (30)$$

where  $M_i$  denotes the model trained by client  $i$ , and  $D_i$  represents the local dataset associated with client  $i$ .

Since training occurs locally, raw vehicle records remain private and are never transmitted to external entities.

#### E. Local Prediction Generation

Following model training, each client generates predictions using the testing dataset. For a given client  $i$ , the prediction output is represented as:

$$\hat{y}_i = M_i(X), \quad (31)$$

where  $M_i$  represents the local model,  $X$  denotes the input feature matrix, and  $\hat{y}_i$  represents the prediction generated by client  $i$ .

Each client independently contributes its prediction estimates to the aggregation process. This decentralized prediction mechanism forms the foundation of the federated learning framework.

#### F. Weighted Aggregation Strategy

The proposed framework employs a weighted aggregation mechanism to combine predictions generated by individual clients. Instead of assigning equal importance to all clients, aggregation weights are computed according to client dataset size.

Let  $|D_i|$  represent the number of samples contained within client  $i$ . The aggregation weight assigned to client  $i$  is calculated as:

$$w_i = \frac{|D_i|}{\sum_{j=1}^N |D_j|}, \quad (32)$$

where  $w_i$  denotes the aggregation weight, and  $N$  denotes the total number of clients. This weighting strategy ensures that larger clients contribute proportionally more to the final prediction.

The global federated prediction is then computed as:

$$\hat{y} = \sum_{i=1}^N w_i \hat{y}_i, \quad (33)$$

where  $\hat{y}_i$  represents the prediction generated by client  $i$ , and  $\hat{y}$  denotes the final federated prediction.

This aggregation mechanism resembles the principles of Federated Averaging (FedAvg) while operating directly on prediction outputs.

### G. Privacy Preservation Mechanism

One of the primary motivations for adopting federated learning is the preservation of user privacy. In the proposed framework:

- Raw vehicle records remain local.
- Vehicle telemetry is not transmitted.
- Battery information is not shared.
- Driver behavior data remains private.
- Environmental observations remain local.

Only model outputs contribute to the aggregation process. As a result, the federated architecture significantly reduces the privacy risks associated with centralized data collection while maintaining collaborative learning capabilities. This characteristic is particularly important in modern intelligent transportation systems where regulatory requirements and user privacy concerns continue to increase.

### H. Federated Learning Workflow

The complete federated learning process can be summarized as follows:

- 1) **Data Partitioning:** Vehicle records are divided among twenty federated clients.
- 2) **Local Feature Engineering:** Physics-informed features are generated independently at each client.
- 3) **Local Model Training:** Each client trains a LightGBM prediction model using local data.
- 4) **Local Prediction Generation:** Local models generate energy consumption predictions.
- 5) **Weighted Aggregation:** Client predictions are combined according to sample-size-based weights.
- 6) **Global Prediction:** The final federated prediction is produced.
- 7) **Performance Evaluation:** Prediction accuracy is evaluated using standard regression metrics.

This workflow enables privacy-preserving learning without sacrificing predictive performance.

### I. Comparative Advantages of Federated Learning

Compared with centralized learning, the federated framework offers several important advantages.

- **Privacy Preservation:** Vehicle data remains local and is never shared.
- **Scalability:** New clients can be incorporated without restructuring the overall architecture.
- **Reduced Data Transfer:** Large-scale transportation datasets do not need to be transmitted to a central repository.
- **Regulatory Compliance:** The framework aligns with privacy-aware data management principles.
- **Realistic Deployment:** The architecture closely resembles real-world distributed transportation environments.

These characteristics make federated learning an attractive solution for future intelligent mobility systems.

### J. Federated Learning Performance

The federated framework was evaluated using the same testing dataset employed for centralized model assessment. Experimental results demonstrated that the federated model achieved:

$$\text{MAE} = 0.4844. \quad (34)$$

Compared with the centralized hybrid model:

$$\text{MAE} = 0.4629, \quad (35)$$

the federated framework exhibited only a marginal reduction in predictive accuracy.

This observation indicates that privacy-preserving distributed learning can be achieved with minimal performance degradation. The results further demonstrate the feasibility of deploying federated energy prediction systems across large-scale electric vehicle networks.

### K. Summary

This section presented the federated learning framework developed for privacy-preserving electric vehicle energy consumption prediction. The proposed architecture partitions vehicle records across twenty distributed clients, performs local LightGBM model training, and combines client predictions using weighted aggregation. By eliminating the need for centralized data collection, the framework preserves privacy while maintaining predictive performance comparable to centralized learning. Experimental evaluation confirms the effectiveness of the federated approach and highlights its suitability for large-scale intelligent transportation applications.

TABLE III  
EXPERIMENTAL ENVIRONMENT

| Component    | Purpose                             |
|--------------|-------------------------------------|
| Python       | Core programming language           |
| Pandas       | Data preprocessing and manipulation |
| NumPy        | Numerical computation               |
| LightGBM     | Machine learning model development  |
| Scikit-Learn | Model evaluation and data splitting |
| Joblib       | Model persistence                   |
| Matplotlib   | Visualization and analysis          |
| SUMO         | Real-time traffic simulation        |
| Streamlit    | Dashboard development               |

## VIII. EXPERIMENTAL SETUP

### A. Overview

The experimental setup was designed to evaluate the effectiveness of the proposed Physics-Informed Federated Learning Framework for electric vehicle energy consumption prediction. The experiments investigate the contribution of physics-informed feature engineering, machine learning-based prediction, and federated learning toward improving predictive performance while preserving privacy.

To ensure a comprehensive evaluation, multiple model configurations were developed and compared using identical datasets, training procedures, and evaluation metrics. The experimental framework enables assessment of both centralized and federated learning paradigms while quantifying the impact of physics-based feature engineering on prediction accuracy.

The complete experimental workflow consists of dataset preparation, feature engineering, model development, federated learning implementation, performance evaluation, and real-time deployment validation through traffic simulation.

### B. Experimental Environment

All experiments were conducted using the Python programming language and a collection of open-source machine learning and data analysis libraries. The software environment utilized in this study is summarized in Table III.

The combination of these tools provides a complete ecosystem for data processing, machine learning, simulation, and visualization.

### C. Dataset Configuration

The experiments utilize an electric vehicle operational dataset consisting of 5,000 observations. The dataset contains vehicle telemetry information, battery characteristics, environmental variables, and road-condition attributes.

The target variable is:

$$\text{Energy\_Consumption\_kWh}, \quad (36)$$

which represents the energy consumed by the vehicle during a particular operating interval.

The dataset includes:

- Vehicle dynamics variables
- Battery-related variables
- Environmental variables
- Road-condition variables
- Travel-related variables

This diverse feature set enables comprehensive modeling of vehicle energy consumption behavior.

### D. Training and Testing Strategy

To evaluate model generalization capability, the dataset was partitioned into training and testing subsets using an 80:20 ratio.

The training dataset contains:

$$4000 \quad (37)$$

samples. The testing dataset contains:

$$1000 \quad (38)$$

samples.

The partitioning process was performed using:

$$\text{Random State} = 42 \quad (39)$$

TABLE IV  
PHYSICS-INFORMED FEATURE CONFIGURATION

| Feature              | Description              |
|----------------------|--------------------------|
| $P_{\text{drag}}$    | Aerodynamic drag power   |
| $P_{\text{roll}}$    | Rolling resistance power |
| $P_{\text{grade}}$   | Grade resistance power   |
| $P_{\text{inertia}}$ | Inertial power           |
| $P_{\text{total}}$   | Total vehicle power      |

to ensure experimental reproducibility.

The testing dataset remained completely unseen during model training and was used exclusively for performance evaluation. This methodology ensures unbiased assessment of model performance.

#### E. Model Configurations

To investigate the effectiveness of different feature combinations, four centralized prediction models and one federated model were evaluated.

1) *Model A: Full Raw Features*: Model A utilizes all original dataset variables except:

- Vehicle\_ID
- Timestamp
- Target Variable

This configuration serves as the baseline machine learning model. The objective is to determine prediction performance using only raw operational information.

2) *Model B: Reduced Feature Model*: Model B removes several vehicle dynamics and driving-related variables:

- Speed\_kmh
- Acceleration\_ms2
- Slope\_%
- Distance\_Travelled\_km
- Driving\_Mode
- Road\_Type
- Traffic\_Condition
- Wind\_Speed\_ms

This configuration evaluates the importance of driving-related variables within the prediction process.

3) *Model C: Physics-Only Model*: Model C utilizes the generated physics-informed features while excluding key raw driving variables. The objective is to evaluate the predictive capability of physics-based information independently of raw telemetry measurements. This configuration provides insight into the contribution of vehicle dynamics equations toward energy prediction.

4) *Model D: Hybrid Physics-Informed Model*: Model D combines:

- Raw operational variables
- Physics-informed variables

This hybrid configuration represents the proposed centralized framework. The objective is to determine whether combining physical knowledge and machine learning improves predictive performance.

5) *Federated Model*: The federated model utilizes the same hybrid feature representation as Model D but distributes training across multiple clients. The federated architecture evaluates privacy-preserving distributed learning while maintaining predictive accuracy.

#### F. Physics-Informed Feature Configuration

Five physics-based variables were generated using vehicle dynamics equations, as summarized in Table IV.

These variables were computed using speed, acceleration, road slope, and vehicle mass information. The generated features were subsequently integrated with the original dataset to create the hybrid feature representation used by the proposed framework.

#### G. Centralized Model Configuration

The centralized prediction model employs a Light Gradient Boosting Machine (LightGBM) regressor. The selected hyperparameters are shown in Table V.

LightGBM was selected because of its ability to efficiently model nonlinear relationships within structured tabular datasets while maintaining low computational complexity.

TABLE V  
CENTRALIZED LIGHTGBM HYPERPARAMETERS

| Parameter       | Value              |
|-----------------|--------------------|
| Algorithm       | LightGBM Regressor |
| Number of Trees | 300                |
| Learning Rate   | 0.05               |
| Maximum Depth   | 6                  |

TABLE VI  
FEDERATED LEARNING CONFIGURATION

| Parameter          | Value                |
|--------------------|----------------------|
| Number of Clients  | 20                   |
| Partition Strategy | Vehicle-Based        |
| Aggregation Method | Weighted Aggregation |
| Local Model        | LightGBM             |

TABLE VII  
LOCAL CLIENT HYPERPARAMETERS

| Parameter       | Value              |
|-----------------|--------------------|
| Algorithm       | LightGBM Regressor |
| Number of Trees | 200                |
| Learning Rate   | 0.05               |
| Maximum Depth   | 6                  |

#### H. Federated Learning Configuration

The federated learning framework partitions the dataset according to vehicle identifiers. The partitioning strategy ensures that records associated with a specific vehicle remain within the same client. The federated configuration is summarized in Table VI.

Each client independently trains a local model using its assigned subset of vehicle records. The aggregation weight assigned to each client is proportional to its dataset size. This strategy enables privacy-preserving collaborative learning without requiring raw data sharing.

#### I. Local Client Configuration

Each federated client trains an independent LightGBM model using the hyperparameters shown in Table VII. The local configuration is intentionally lightweight to support scalable federated deployment.

#### J. Evaluation Metrics

Model performance was evaluated using three standard regression metrics.

1) *Mean Absolute Error*: Mean Absolute Error (MAE) measures the average magnitude of prediction errors:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (40)$$

Lower MAE values indicate better prediction performance.

2) *Root Mean Square Error*: Root Mean Square Error (RMSE) measures the square root of the average squared prediction error:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (41)$$

RMSE penalizes large prediction errors more heavily than MAE.

3) *Coefficient of Determination*: The coefficient of determination ( $R^2$ ) evaluates the proportion of variance explained by the prediction model:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (42)$$

Higher values indicate better predictive capability.

### K. Visualization and Performance Analysis

Several visualization techniques were employed to analyze model behavior and prediction quality. These include:

- Actual vs. Predicted Scatter Plot
- Error Distribution Histogram
- Feature Importance Analysis
- MAE Comparison Graph
- Vehicle-Level Energy Analytics
- Efficiency Trend Visualization
- Prediction Error Analysis

These visualizations provide comprehensive insight into model performance and prediction characteristics.

### L. SUMO-Based Validation

To evaluate real-world deployment capability, the trained prediction model was integrated with the Simulation of Urban Mobility (SUMO) platform.

During simulation:

- 1) Vehicle movement data is generated.
- 2) Vehicle speed and acceleration are extracted.
- 3) Physics-informed features are computed.
- 4) Energy consumption predictions are generated.
- 5) Vehicle-level analytics are displayed.

This integration enables evaluation of the framework under realistic transportation conditions.

### M. Experimental Objectives

The experimental study seeks to answer the following research questions:

- **RQ1:** Does physics-informed feature engineering improve energy prediction accuracy?
- **RQ2:** How does the hybrid model compare with raw-feature and physics-only approaches?
- **RQ3:** Can federated learning preserve prediction performance while enhancing privacy?
- **RQ4:** Is the proposed framework suitable for real-time deployment within transportation simulation environments?

The results presented in the following section provide detailed answers to these questions.

### N. Summary

This section presented the experimental setup used to evaluate the proposed Physics-Informed Federated Learning Framework. A comprehensive evaluation methodology was established involving multiple prediction models, federated learning configurations, physics-informed feature engineering, standardized evaluation metrics, and SUMO-based deployment validation. The resulting framework enables rigorous assessment of predictive accuracy, privacy preservation, scalability, and real-time applicability.

## IX. RESULTS AND DISCUSSION

### A. Overview

This section presents the experimental results obtained from the proposed Physics-Informed Federated Learning Framework for electric vehicle energy consumption prediction. The objective of the evaluation is to analyze the effectiveness of physics-informed feature engineering, assess the performance of federated learning, and investigate the contribution of different feature configurations toward prediction accuracy.

To provide a comprehensive assessment, five different model configurations were evaluated:

- 1) Model A – Full Raw Features
- 2) Model B – Reduced Feature Model
- 3) Model C – Physics-Only Model
- 4) Model D – Hybrid Physics-Informed Model
- 5) Federated Hybrid Model

The results are analyzed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination ( $R^2$ ).

TABLE VIII  
COMPARATIVE PERFORMANCE OF EVALUATED MODELS

| Model           | Description                   | MAE (kWh) |
|-----------------|-------------------------------|-----------|
| Model A         | Full Raw Features             | 0.4680    |
| Model B         | Reduced Features              | 1.7937    |
| Model C         | Physics-Only Features         | 1.0133    |
| Model D         | Hybrid Physics-Informed Model | 0.4629    |
| Federated Model | Federated Hybrid Model        | 0.4844    |

TABLE IX  
CENTRALIZED HYBRID MODEL PERFORMANCE

| Metric | Value  |
|--------|--------|
| MAE    | 0.4629 |
| RMSE   | 0.5744 |
| $R^2$  | 0.9320 |

### B. Comparative Analysis of Model Configurations

A primary objective of this study was to evaluate the contribution of different feature groups toward electric vehicle energy prediction. Table VIII summarizes the performance obtained by each model configuration.

The results reveal several important observations. First, Model B produced the highest prediction error among all evaluated configurations. The removal of driving-related variables such as speed, acceleration, road slope, travel distance, and traffic information resulted in a substantial degradation in performance. This observation confirms that vehicle dynamics variables are critical predictors of electric vehicle energy consumption.

Second, the Physics-Only Model achieved significantly better performance than the Reduced Feature Model. This result demonstrates that physics-derived variables contain meaningful information regarding vehicle energy demand even when several raw driving variables are excluded.

Most importantly, the Hybrid Physics-Informed Model achieved the lowest prediction error of all evaluated approaches. The integration of raw telemetry information with physics-informed variables enabled the model to exploit both statistical relationships and physical principles simultaneously.

These findings confirm that the proposed hybrid feature representation provides the most comprehensive description of vehicle energy behavior.

### C. Centralized Model Performance

The Hybrid Physics-Informed Model (Model D) was selected as the primary centralized prediction model because it achieved the best overall performance. The obtained results are summarized in Table IX.

The model achieved a Mean Absolute Error of 0.4629 kWh, indicating that the average prediction deviation from actual energy consumption values is less than half a kilowatt-hour.

The Root Mean Square Error of 0.5744 kWh further demonstrates the model's ability to maintain low prediction errors across diverse operating conditions.

Most notably, the model achieved an  $R^2$  score of 0.9320, indicating that approximately 93.2% of the variance in vehicle energy consumption is explained by the proposed framework.

This result demonstrates strong predictive capability and highlights the effectiveness of combining physics-informed feature engineering with gradient boosting techniques.

### D. Effectiveness of Physics-Informed Features

One of the primary research questions addressed in this study concerns the contribution of physics-informed variables toward prediction performance. A comparison between Models A, C, and D provides valuable insight into the effectiveness of the proposed feature engineering strategy.

Model A utilized only raw telemetry features and achieved an MAE of 0.4680 kWh. Model C utilized physics-informed features and achieved an MAE of 1.0133 kWh. Model D combined both feature groups and achieved an MAE of 0.4629 kWh.

These results indicate that neither raw features nor physics-informed variables alone provide the optimal solution. Instead, the best performance is obtained when both information sources are combined.

The improvement achieved by the hybrid model suggests that physics-informed variables provide complementary information not fully captured by raw telemetry measurements. By explicitly modeling vehicle energy dynamics, the generated features help guide the machine learning model toward more physically meaningful predictions.

This finding validates the central hypothesis of this research and demonstrates the value of integrating domain knowledge into data-driven prediction systems.



TABLE X  
MOST INFLUENTIAL FEATURES

| Rank | Feature               |
|------|-----------------------|
| 1    | Battery_State_%       |
| 2    | Distance_Travelled_km |
| 3    | Humidity_%            |
| 4    | Battery_Temperature_C |
| 5    | Speed_kmh             |
| 6    | Wind_Speed_ms         |
| 7    | Battery_Voltage_V     |
| 8    | $P_{\text{grade}}$    |
| 9    | Tire_Pressure_psi     |
| 10   | Acceleration_ms2      |

#### E. Actual versus Predicted Analysis

To evaluate the quality of individual predictions, actual energy consumption values were compared against model predictions. The resulting Actual vs. Predicted plot exhibited a strong linear relationship between observed and estimated energy consumption values.

Most observations were concentrated near the ideal prediction line, indicating close agreement between predicted and actual values. The strong alignment observed in the scatter distribution is consistent with the high  $R^2$  score obtained during evaluation and demonstrates the ability of the model to generalize effectively across different operating conditions.

The results further suggest that the model successfully captures the dominant factors influencing energy consumption and produces stable predictions throughout the dataset.

#### F. Error Distribution Analysis

Error analysis provides important insights regarding model reliability and robustness. The error distribution histogram revealed that prediction errors are concentrated around zero, indicating the absence of significant systematic bias.

Most prediction errors fall within a relatively narrow range, suggesting that the model maintains consistent performance across diverse operating conditions. The near-symmetric error distribution further indicates that the model does not consistently overestimate or underestimate vehicle energy consumption.

Such characteristics are desirable in practical deployment scenarios because they reduce the likelihood of persistent prediction inaccuracies.

#### G. Feature Importance Analysis

Feature importance analysis was conducted to identify the variables that contribute most strongly to prediction performance. The ten most influential variables identified by the LightGBM model are summarized in Table X.

Several important observations emerge from this analysis. Battery-related variables occupy multiple top-ranking positions, highlighting the strong relationship between battery condition and energy consumption behavior.

Vehicle speed and acceleration also appear among the most influential variables, reflecting the importance of driving behavior and vehicle dynamics.

Notably, the physics-informed variable  $P_{\text{grade}}$  appears among the top predictors. This observation confirms that the generated physics features contribute meaningful information to the prediction process.

The presence of both raw and physics-derived variables among the most influential features further supports the effectiveness of the hybrid modeling strategy.

#### H. Federated Learning Performance

The federated learning framework was evaluated to determine whether privacy-preserving distributed learning could maintain prediction performance comparable to centralized learning. The federated model achieved:

$$\text{MAE} = 0.4844. \quad (43)$$

Compared with the centralized hybrid model:

$$\text{MAE} = 0.4629, \quad (44)$$

the performance difference is relatively small.

The increase in prediction error is approximately:

$$4.64\%. \quad (45)$$

TABLE XI  
EXAMPLE PREDICTION

| Parameter                    | Value     |
|------------------------------|-----------|
| Actual Energy Consumption    | 8.256 kWh |
| Predicted Energy Consumption | 7.466 kWh |
| Absolute Error               | 0.790 kWh |

This modest performance reduction demonstrates that federated learning can preserve most of the predictive capability of centralized models while eliminating the need for centralized data collection.

From a practical perspective, this trade-off is highly favorable because it enables privacy-preserving learning without substantially compromising accuracy.

#### I. Sample Prediction Analysis

A sample prediction generated by the proposed framework is shown in Table XI.

The relatively small prediction error observed in this example demonstrates the model's ability to estimate energy consumption with reasonable precision. Such prediction accuracy is particularly valuable in applications involving route planning, charging optimization, and fleet energy management.

#### J. Discussion

The experimental results demonstrate the effectiveness of combining physics-informed feature engineering, machine learning, and federated learning within a unified prediction framework. Several important conclusions can be drawn from the analysis.

First, vehicle dynamics variables play a critical role in energy prediction. The substantial performance degradation observed in the Reduced Feature Model highlights the importance of speed, acceleration, distance, and road-related information.

Second, physics-informed features provide meaningful complementary information that improves model performance beyond what can be achieved using raw telemetry alone. The superior performance of the Hybrid Physics-Informed Model confirms the value of integrating engineering knowledge into machine learning systems.

Third, federated learning successfully preserves predictive capability while addressing privacy concerns. The small difference between centralized and federated performance demonstrates that privacy-preserving distributed learning is feasible for electric vehicle energy prediction applications.

Finally, the combination of high predictive accuracy, physical interpretability, privacy preservation, and simulation-based deployment demonstrates the practicality of the proposed framework for real-world intelligent transportation systems.

#### K. Key Findings

The major findings of this study can be summarized as follows:

- 1) The Hybrid Physics-Informed Model achieved the best overall performance with an MAE of 0.4629 kWh.
- 2) The proposed framework achieved an  $R^2$  score of 0.9320, indicating excellent predictive capability.
- 3) Physics-informed features improved prediction accuracy by providing physically meaningful representations of vehicle energy dynamics.
- 4) Federated learning preserved model performance while enabling privacy-preserving distributed training.
- 5) The performance difference between centralized and federated learning remained below 5%, demonstrating the practicality of privacy-aware energy prediction systems.
- 6) The proposed framework successfully integrates machine learning, physics-based modeling, federated learning, SUMO simulation, and dashboard analytics within a unified architecture.

These findings collectively demonstrate the effectiveness and practical applicability of the proposed Physics-Informed Federated Learning Framework for electric vehicle energy consumption prediction.

### X. SUMO-BASED REAL-TIME DEPLOYMENT FRAMEWORK

#### A. Introduction

While offline evaluation provides valuable insights regarding model accuracy, practical intelligent transportation systems require prediction frameworks capable of operating under real-time traffic conditions. Consequently, the proposed Physics-Informed Federated Learning Framework was integrated with the Simulation of Urban Mobility (SUMO) platform to evaluate deployment feasibility within a dynamic transportation environment.

SUMO is an open-source microscopic traffic simulator widely used for modeling vehicle movement, traffic interactions, route planning, and intelligent transportation systems. The platform provides detailed vehicle-level information at every simulation step, making it suitable for evaluating real-time energy prediction systems.

The objective of the SUMO integration is to demonstrate that the proposed framework can continuously estimate electric vehicle energy consumption while vehicles move through a simulated transportation network.

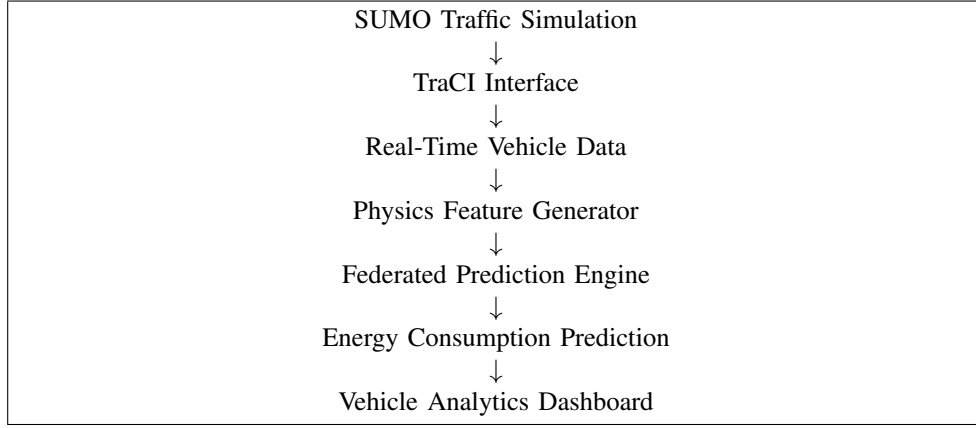


Fig. 4. SUMO integration architecture.

TABLE XII  
REAL-TIME VEHICLE DATA ACQUIRED FROM SUMO

| Parameter         | Description                |
|-------------------|----------------------------|
| Vehicle ID        | Unique vehicle identifier  |
| Speed             | Current vehicle speed      |
| Acceleration      | Instantaneous acceleration |
| Position          | Vehicle location           |
| Route Information | Active route details       |

### B. SUMO Integration Architecture

The real-time deployment framework consists of five major components:

- 1) SUMO Traffic Simulation Engine
- 2) TraCI Communication Interface
- 3) Physics-Informed Feature Generation Module
- 4) Federated Prediction Engine
- 5) Real-Time Monitoring and Analytics Layer

The overall deployment architecture is illustrated in Fig. 4.

The architecture enables continuous communication between the traffic simulator and the prediction framework.

### C. Real-Time Vehicle Data Acquisition

The Traffic Control Interface (TraCI) serves as the communication bridge between SUMO and the machine learning framework. At every simulation step, the system extracts operational information from all active vehicles.

The retrieved parameters are summarized in Table XII.

Among these variables, speed and acceleration are directly utilized within the energy prediction pipeline. The real-time acquisition process enables continuous monitoring of vehicle behavior throughout the simulation.

### D. Dynamic Feature Construction

The prediction framework requires a complete feature vector at every simulation step. Since SUMO primarily provides vehicle movement information, additional variables are generated using predefined operational assumptions.

The assigned deployment parameters are summarized in Table XIII.

These variables are combined with real-time speed and acceleration measurements to create the prediction input vector. The resulting feature representation closely matches the structure used during model training.

### E. Physics-Informed Feature Generation

Following data acquisition, the physics-informed feature engineering module computes vehicle power components in real time. The generated features include aerodynamic drag power, rolling resistance power, grade resistance power, inertial power,

TABLE XIII  
ASSIGNED VALUES FOR DYNAMIC FEATURE CONSTRUCTION

| Variable            | Assigned Value |
|---------------------|----------------|
| Battery State       | 60%            |
| Battery Voltage     | 350 V          |
| Battery Temperature | 30°C           |
| Ambient Temperature | 30°C           |
| Humidity            | 60%            |
| Wind Speed          | 2 m/s          |
| Tire Pressure       | 32 psi         |
| Vehicle Weight      | 1700 kg        |
| Road Slope          | 0%             |

and total vehicle power:

$$P_{\text{drag}} = \frac{1}{2} \rho C_d A v^3, \quad (46)$$

$$P_{\text{roll}} = mg C_r v, \quad (47)$$

$$P_{\text{grade}} = mg S v, \quad (48)$$

$$P_{\text{inertia}} = m a v, \quad (49)$$

$$P_{\text{total}} = P_{\text{drag}} + P_{\text{roll}} + P_{\text{grade}} + P_{\text{inertia}}. \quad (50)$$

These features provide physically meaningful representations of the instantaneous energy demand experienced by each vehicle. The generated variables are subsequently combined with operational measurements to create the hybrid feature vector used by the prediction model.

#### F. Federated Prediction During Simulation

Once the hybrid feature vector has been generated, predictions are obtained using the trained federated learning framework. Each local model independently estimates vehicle energy consumption:

$$\hat{y}_i = M_i(X), \quad (51)$$

where  $M_i$  denotes the local client model, and  $X$  represents the hybrid feature vector.

The final energy prediction is obtained through weighted aggregation:

$$\hat{y} = \sum_{i=1}^N w_i \hat{y}_i, \quad (52)$$

where  $N = 20$  federated clients, and  $w_i$  denotes the aggregation weight.

This process enables real-time energy estimation while preserving the federated learning architecture developed during training.

#### G. Energy Efficiency Estimation

In addition to energy prediction, the framework computes vehicle energy efficiency. Energy efficiency is calculated as:

$$\text{Efficiency} = \frac{\text{Energy}}{\text{Distance}}, \quad (53)$$

where Energy represents predicted energy consumption, and Distance represents the traveled distance.

The efficiency metric provides additional insights regarding vehicle operating behavior and route performance. Vehicles exhibiting lower efficiency values may indicate aggressive driving patterns, unfavorable traffic conditions, or inefficient route selection.

#### H. Battery State Monitoring

To simulate realistic electric vehicle operation, a battery state monitoring module was incorporated. Each vehicle begins with an initial battery state:

$$\text{SOC}_{\text{initial}} = 100\%. \quad (54)$$

The battery state is updated according to predicted energy consumption:

$$\text{SOC}_{t+1} = \text{SOC}_t - (\text{Energy} \times 0.01), \quad (55)$$

where  $\text{SOC}_t$  represents the current battery state, and Energy denotes predicted energy consumption.

This mechanism enables visualization of battery depletion during simulation and provides a realistic representation of electric vehicle operation.

TABLE XIV  
VEHICLE ENERGY CLASSIFICATION

| Category                    | Energy Range               | Vehicle Color |
|-----------------------------|----------------------------|---------------|
| Low Energy Consumption      | Energy < 5                 | Green         |
| Moderate Energy Consumption | $5 \leq \text{Energy} < 8$ | Yellow        |
| High Energy Consumption     | Energy $\geq 8$            | Red           |

### I. Vehicle Energy Classification

The framework incorporates a visual classification mechanism that identifies vehicles according to their energy consumption levels. Three categories are defined in Table XIV.

This visual representation enables immediate identification of high-energy vehicles within the traffic simulation environment.

### J. Vehicle-Level Analytics

The deployment framework continuously maintains vehicle-level statistics throughout the simulation. For each active vehicle, the following information is recorded:

- Vehicle ID
- Speed
- Predicted Energy Consumption
- Energy Efficiency
- Battery State

These metrics enable detailed analysis of vehicle performance and facilitate identification of energy-intensive operating conditions. The framework also generates rankings of the highest energy-consuming vehicles, allowing transportation planners to identify patterns associated with inefficient operation.

### K. Real-Time Output Generation

During simulation, the framework continuously outputs prediction information for each active vehicle. The generated output includes vehicle ID, speed, predicted energy consumption, energy efficiency, and battery state.

A representative output format is shown below:

```
Vehicle: EV_001
Speed: 42.7 km/h
Energy: 6.214 kWh
Efficiency: 1.483 kWh/km
Battery: 96.4 %
```

This information enables continuous monitoring of fleet-level energy behavior throughout the simulation period.

### L. Significance of SUMO Integration

The integration of the proposed framework with SUMO provides several important benefits. First, it demonstrates that the prediction model can operate under real-time transportation conditions rather than only within offline datasets. Second, it enables evaluation of vehicle energy behavior under diverse traffic scenarios. Third, it provides a deployment environment for testing federated prediction systems before real-world implementation. Finally, the integration bridges the gap between machine learning model development and intelligent transportation system deployment.

The successful deployment of the framework within SUMO confirms its applicability to future smart mobility ecosystems and real-time electric vehicle energy management systems.

## XI. DASHBOARD ANALYTICS AND VISUALIZATION FRAMEWORK

### A. Introduction

While predictive accuracy remains a fundamental objective of machine learning systems, practical deployment requires mechanisms that enable users to interpret, monitor, and utilize prediction outputs effectively. Consequently, an interactive analytics dashboard was developed as the final layer of the proposed Physics-Informed Federated Learning Framework.

The dashboard serves as a decision-support platform that transforms prediction outputs into intuitive visual representations. Through interactive analytics, users can evaluate model performance, investigate vehicle-level energy behavior, identify operational trends, and monitor real-time prediction outcomes.

The dashboard was developed using the Streamlit framework and integrated directly with the trained prediction model and simulation outputs. This integration enables both offline analysis and real-time monitoring of electric vehicle energy consumption.

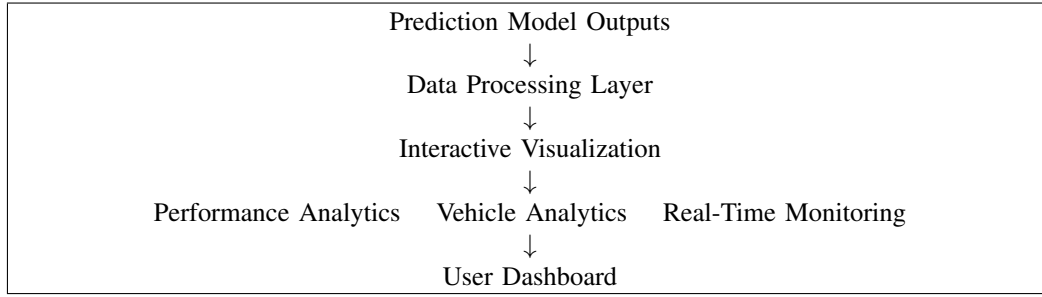


Fig. 5. Dashboard analytics architecture.

### B. Dashboard Architecture

The dashboard architecture consists of four major modules:

- 1) Data Analytics Module
- 2) Model Performance Evaluation Module
- 3) Vehicle-Level Energy Analytics Module
- 4) Real-Time Monitoring Module

The overall dashboard workflow is illustrated in Fig. 5.

The architecture enables continuous interaction between prediction outputs and visualization components.

### C. Dataset Analytics Module

The first dashboard component provides a comprehensive overview of the dataset utilized for model development. This module displays:

- Dataset dimensions
- Feature information
- Statistical summaries
- Distribution characteristics

The objective is to provide users with an understanding of the underlying data structure before model evaluation. Dataset analytics facilitate identification of data quality issues and support exploratory analysis of energy-related variables.

### D. Model Performance Analytics

A dedicated performance evaluation module was developed to assess prediction quality. This module presents the primary regression metrics:

$$\text{MAE} = 0.4629, \quad (56)$$

$$\text{RMSE} = 0.5744, \quad (57)$$

$$R^2 = 0.9320. \quad (58)$$

These metrics provide a quantitative assessment of model accuracy and enable comparison between different prediction approaches. The dashboard displays these values using metric cards to facilitate rapid interpretation.

### E. Actual versus Predicted Visualization

One of the most important dashboard components is the Actual versus Predicted analysis. This visualization compares observed energy consumption values with model predictions. Ideally, all observations should lie along the diagonal line representing perfect prediction.

The dashboard reveals a strong agreement between actual and predicted values, indicating that the proposed framework successfully captures the dominant factors influencing electric vehicle energy consumption. The close alignment observed within the scatter distribution is consistent with the high coefficient of determination achieved during evaluation.

This visualization provides immediate confirmation of model effectiveness and prediction reliability.

### F. Error Distribution Analysis

Understanding prediction errors is essential for evaluating model robustness. The dashboard includes an error distribution histogram that illustrates the frequency of prediction errors across the testing dataset.

The observed distribution is centered near zero, indicating the absence of substantial systematic bias. The relatively narrow error range demonstrates that the framework maintains stable prediction performance across different operating conditions.

Error analysis enables users to identify unusual prediction behavior and assess model reliability prior to deployment.

### G. Residual Analysis

Residual analysis provides additional insight into prediction quality. Residual values are computed as:

$$\text{Residual} = \text{Actual} - \text{Predicted}. \quad (59)$$

The residual plot visualizes the relationship between prediction errors and predicted energy consumption. A well-performing model should exhibit randomly distributed residuals without obvious patterns.

The dashboard confirms that residuals remain largely centered around zero and do not display strong systematic trends. This observation suggests that the prediction model effectively captures the majority of meaningful relationships present within the dataset.

### H. Vehicle Energy Consumption Distribution

The dashboard includes a vehicle energy distribution module designed to analyze fleet-wide energy consumption patterns. This visualization presents the distribution of energy consumption values across all vehicles.

The analysis enables identification of:

- Low-energy vehicles
- Moderate-energy vehicles
- High-energy vehicles

Such information is particularly valuable for fleet operators seeking to identify inefficient vehicles or unusual operating conditions. The distribution analysis also provides insights regarding the variability of energy demand within transportation systems.

### I. Trip Distance versus Energy Consumption Analysis

Trip distance is one of the most influential factors affecting energy demand. To investigate this relationship, the dashboard incorporates a Trip Distance versus Energy Consumption visualization.

The resulting scatter plot illustrates the correlation between travelled distance and energy usage. A positive relationship is generally observed, indicating that longer trips require greater energy expenditure. However, deviations from the overall trend reveal the influence of additional factors such as traffic conditions, road gradients, weather variables, and driving behavior.

This visualization enables users to better understand the relationship between trip characteristics and vehicle energy demand.

### J. Vehicle-Level Analytics

The dashboard provides detailed vehicle-level analytics that support granular investigation of vehicle behavior. For each vehicle, users can examine:

- Energy consumption
- Vehicle speed
- Battery status
- Travel distance
- Efficiency indicators

Vehicle-level monitoring facilitates identification of energy-intensive operating conditions and supports optimization of fleet management strategies. This capability is particularly useful in large-scale transportation environments involving hundreds or thousands of vehicles.

### K. Energy Efficiency Trend Analysis

Energy efficiency serves as an important indicator of vehicle performance. The dashboard calculates efficiency as:

$$\text{Efficiency} = \frac{\text{Energy Consumption}}{\text{Distance Travelled}}, \quad (60)$$

and visualizes efficiency trends across the dataset.

The resulting analysis provides insight into:

- Vehicle operating efficiency
- Route effectiveness
- Driving behavior
- Energy optimization opportunities

Monitoring efficiency trends enables transportation planners to identify operational practices that reduce energy expenditure and improve sustainability.

### *L. Correlation Heatmap Analysis*

Understanding relationships among variables is critical for interpreting machine learning models. The dashboard includes an interactive correlation heatmap that visualizes pairwise relationships between dataset features.

Strong positive and negative correlations can be identified immediately, allowing users to understand the factors most closely associated with energy consumption. The heatmap also assists in validating feature engineering decisions and identifying potential multicollinearity issues.

As a result, the correlation analysis contributes both to model interpretation and dataset understanding.

### *M. Real-Time Prediction Monitoring*

A distinguishing feature of the dashboard is its ability to display real-time prediction outputs generated during deployment. As vehicles move through the simulation environment, the dashboard continuously updates:

- Vehicle speed
- Predicted energy consumption
- Battery state
- Efficiency metrics

This functionality transforms the dashboard from a static reporting tool into a real-time monitoring platform. The ability to observe prediction outputs dynamically demonstrates the deployment readiness of the proposed framework.

### *N. Practical Significance of Dashboard Analytics*

The dashboard provides several practical benefits beyond conventional model evaluation. First, it improves interpretability by transforming numerical outputs into intuitive visual representations. Second, it enables continuous monitoring of vehicle energy behavior. Third, it supports data-driven decision-making by providing insights into vehicle efficiency, energy consumption trends, and prediction performance. Finally, the dashboard serves as an operational interface that bridges the gap between machine learning research and real-world deployment.

Consequently, the visualization framework represents a critical component of the overall system architecture.

## XII. LIMITATIONS OF THE STUDY

### *A. Dataset Size Limitation*

Although the dataset used in this study contains 5,000 electric vehicle records and captures multiple operational variables, it remains relatively small compared with large-scale industrial datasets collected from commercial electric vehicle fleets. Larger datasets containing millions of observations may reveal additional behavioral patterns and improve model generalization.

### *B. Synthetic and Simulated Environment*

A significant portion of the experimental evaluation was conducted using simulated transportation environments. While SUMO provides realistic traffic behavior and mobility patterns, simulated environments cannot perfectly reproduce all complexities of real-world transportation systems. Factors such as unexpected driver behavior, road incidents, charging station congestion, and extreme weather conditions may influence actual energy consumption differently.

### *C. Simplified Battery Modeling*

The proposed framework utilizes battery-related variables such as state of charge, battery temperature, and battery voltage. However, detailed electrochemical battery degradation processes were not explicitly modeled. Long-term battery aging effects and degradation dynamics may influence energy consumption and prediction accuracy over extended operational periods.

### *D. Communication-Efficient Federated Learning*

The implemented federated learning framework focuses primarily on prediction aggregation and privacy preservation. Communication optimization techniques, adaptive aggregation strategies, and advanced federated optimization algorithms were not explored. These aspects may become increasingly important when scaling the framework to thousands of distributed clients.

### *E. Geographic Generalization*

The developed framework was evaluated using a single dataset configuration and simulation environment. Additional studies involving multiple cities, transportation networks, climatic conditions, and vehicle categories are necessary to fully evaluate geographical robustness and transferability.



#### F. Real-Time Deployment Constraints

Although the framework was successfully integrated with SUMO and dashboard analytics, large-scale deployment in production transportation systems may introduce additional computational and infrastructure challenges. These include network latency, distributed data synchronization, edge-device limitations, and large-scale fleet management requirements.

### XIII. FUTURE WORK

Several promising research directions emerge from this study.

#### A. Real-World Fleet Deployment

Future research can focus on deploying the framework within real electric vehicle fleets. Evaluating the model using operational data collected from commercial vehicles would provide valuable insights regarding practical deployment challenges and model robustness.

#### B. Deep Learning-Based Federated Models

The current framework employs LightGBM due to its strong performance on structured tabular datasets. Future studies may investigate federated deep learning architectures such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Transformer-based networks, and Graph Neural Networks for capturing complex temporal dependencies.

#### C. Advanced Physics-Informed Learning

Future work may incorporate additional vehicle dynamics variables, regenerative braking mechanisms, drivetrain efficiency models, battery degradation equations, and thermal management systems. Such enhancements could further improve physical interpretability and predictive accuracy.

#### D. Vehicle-to-Grid Integration

The rapid growth of Vehicle-to-Grid (V2G) technologies presents new opportunities for intelligent energy management. Future versions of the framework may integrate charging optimization, bidirectional power flow prediction, and grid-support services into the learning architecture.

#### E. Edge Intelligence and IoT Deployment

Future research can investigate deployment on edge devices and Internet of Things platforms. Lightweight federated models running directly on connected vehicles could enable real-time energy prediction with minimal communication overhead.

#### F. Explainable Artificial Intelligence

Explainable AI techniques such as SHAP, LIME, and counterfactual analysis may be incorporated to improve model transparency and facilitate trust among transportation planners, policymakers, and fleet operators.

#### G. Smart City Integration

Future frameworks may integrate electric vehicle energy prediction with broader smart-city infrastructures, including intelligent traffic management systems, renewable energy forecasting, charging infrastructure optimization, and urban mobility planning.

### XIV. CONCLUSION

This study presented a Physics-Informed Federated Learning Framework for Electric Vehicle Energy Consumption Prediction with Real-Time Traffic Simulation and Interactive Analytics. The proposed framework combines vehicle dynamics-based feature engineering, machine learning, federated learning, transportation simulation, and dashboard visualization within a unified architecture.

A comprehensive dataset consisting of electric vehicle operational variables, battery characteristics, environmental conditions, and travel-related information was utilized to develop and evaluate the framework. Five physics-informed variables, namely aerodynamic drag power, rolling resistance power, grade resistance power, inertial power, and total vehicle power, were generated using established vehicle dynamics equations and incorporated into the machine learning pipeline.

Experimental evaluation demonstrated that the hybrid physics-informed model achieved superior performance compared with alternative feature configurations. The proposed centralized model achieved a Mean Absolute Error of 0.4629 kWh, a Root

Mean Square Error of 0.5744 kWh, and a coefficient of determination of 0.9320. These results indicate that the framework successfully captures the dominant factors influencing electric vehicle energy consumption.

To address privacy concerns associated with centralized learning, a federated learning framework consisting of twenty distributed clients was developed. The federated model achieved a Mean Absolute Error of 0.4844 kWh, demonstrating that privacy-preserving distributed learning can maintain predictive performance comparable to centralized approaches with only a minor reduction in accuracy.

The trained model was subsequently integrated with the Simulation of Urban Mobility platform to evaluate real-time deployment feasibility. The integration enabled continuous vehicle monitoring, physics-informed feature generation, federated prediction, battery-state estimation, and energy-efficiency analysis within a realistic transportation environment. An interactive dashboard was further developed to provide visualization, performance monitoring, vehicle-level analytics, and decision-support capabilities.

Overall, the results demonstrate that integrating physics-informed feature engineering with federated learning provides an effective solution for accurate, interpretable, scalable, and privacy-aware electric vehicle energy prediction. The proposed framework contributes toward the development of intelligent transportation systems capable of supporting future smart mobility ecosystems, sustainable energy management strategies, and next-generation connected vehicle infrastructures.

#### REFERENCES

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